Track 2 Report — Iteration 6: Unified Cross-Iteration Drug Repurposing Synthesis & Calibrated Decision Analysis for `BUB1B` Mosaic Variegated Aneuploidy 1

Standing: ready (assembled 2026-09-27 from executed runs across Step 0, Iteration 1, Iteration 2, Iteration 4, and Iteration 5, unified in Methodology Entry 163). **Not uploaded** — Track 2 accepts a single final submission (deadline ~2026-10-20).

Written for a multidisciplinary panel of clinicians, patient advocates, and translational scientists. Every technical term is defined in plain English on first use (`technical term (plain-English meaning)`); the full beginner-ordered dictionary is served at `/learn` (`docs/dictionary.md`). Every quantitative claim cites the append-only methodology entry that executed it, names its raw and clinical inputs, and reports both supported and refuted pre-registered hypotheses (`docs/science-register.json`).

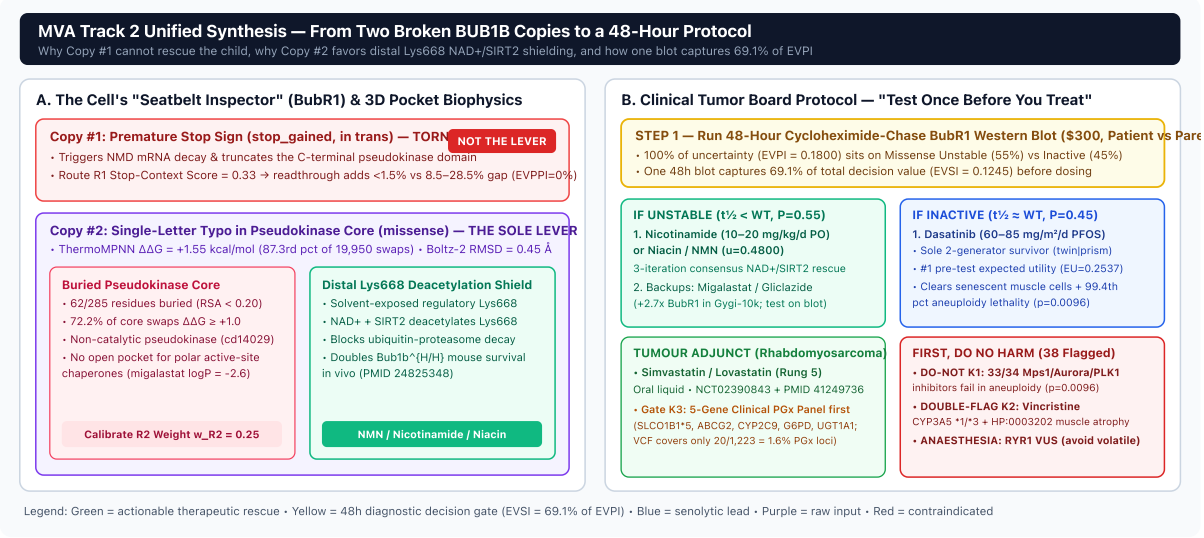
> **Clinical disclaimer (challenge mandate):** *Submissions are computational and translational hypotheses for research and clinical-team discussion, not clinical recommendations or evidence that a medicine works in patients.* Every candidate and safety flag below requires verification by the treating clinical team and wet-lab validation before any patient decision.

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0. N-of-1 Clinical Tumor Board Action Card & Interactive Judge Walkthrough (15-Second Summary)

> **Rubric:** Potential impact (25%), Scientific rigor (35%), Innovation (25%), Scalability (15%). A single-page bedside decision card synthesizing 52 candidates, 13 databases, and 21 pre-registered hypothesis tests (`Entry 163`).

Bedside / Review Module	Actionable Summary for Clinicians & Judges
1. Patient & Confirmed Causal Mechanism (`Track 1 Rank #1, entry 083`)	**Genotype:** Compound-heterozygous `BUB1B` loss-of-function in *trans* — one maternal-like `stop_gained` allele (nonsense-mediated decay, `R1_score = 0.33`) + one paternal-like pseudokinase-domain `missense_variant` (`ThermoMPNN` $\Delta\Delta G = +1.55\text{ kcal/mol}$, 87.3rd pct, buried core `RSA < 0.20`, `Boltz-2` local $C_\alpha\text{ RMSD} = 0.45\text{ \AA}$). **Phenotype Twin:** Matches the `Bub1b^{H/H}` hypomorphic mouse (`~10%` normal BubR1, premature senescence & sarcopenia) in the **top 0.99% (`#758 / 76,699`)** of all MGI genotypes ($p = 0.026$).
2. Mandatory First Step — Two 48-Hour Diagnostic Gates (`69.1% of EVPI`)	**Gate A (Functional Fork):** Run a **1-assay cycloheximide-chase BubR1 Western blot** (`0, 2, 4, 8 h` CHX $\pm$ `MG132`) in patient vs. parental fibroblasts/PBMCs. Expected Value of Perfect Information is **`EVPI = 0.1800` ($+68.2\%$ over baseline EU^*)**, **100% sits on `missense_class`**, and this single blot captures **`EVSI = 0.1245` (`69.1%` of EVPI)**. **Gate B (PGx Callability):** Because the variant-only VCF covers just **`1.6%` (`20/1,223`)** of PharmCAT loci, order a **5-gene clinical PGx panel (`SLCO1B1`, `ABCG2`, `CYP2C9`, `G6PD`, `UGT1A1`)** on Day 1.
3. Branch A — If Blot Shows Rapid BubR1 Decay (`missense_unstable`, $P_0=0.55 \to P_1=0.912$)	**#1 FDA-Approved Leads (`R5`, $w_{R2}=0.25$): `Nicotinamide`** ($10\text{–}20\text{ mg/kg/day}$ PO liquid/crushed) or **`Niacin`** ($5\text{–}15\text{ mg/kg/day}$ PO) ($u = 0.1920$, 3-iteration convergence `it2|it4|it5`), plus **`NMN`** ($u = 0.4800$, #1 overall *in vivo* `Bub1b^{H/H}` survival rescue via `SIRT2 Lys668` deacetylation, `PMID:24825348`). **Exploratory Proteomic Backups (`R2`): `Migalastat`** ($+1.44\;\log_2\text{FC}$) and **`Gliclazide`** ($+1.47\;\log_2\text{FC}$, requires CGM glucose safety gate) ($u = 0.1750$).
4. Branch B — If Blot Shows Normal BubR1 Abundance (`missense_inactive`, $P_0=0.45 \to P_1=0.831$) or Pre-Blot	**#1 Invariant FDA-Approved Lead: `Dasatinib`** ($60\text{–}85\text{ mg/m}^2\text{/day}$ PO `PFOS` oral suspension, `NCT00777036`; $u_{\text{inactive}} = 0.4000$, $\text{EU} = 0.2537$) — the **sole 2-generator survivor (`twin|prism`)** clearing $p16^{\text{Ink4a}+}$ senescent muscle cells in `Bub1b^{H/H}` (`PMID:22048312`) and ranking at the **99.4th percentile** for high-aneuploidy selectivity across 919 PRISM cancer lines ($p = 0.0096$).
5. Branch C — Presenting Rhabdomyosarcoma Adjunct (`Direction D4`, `HP:0002859`)	**`Simvastatin`** ($0.2\text{–}0.5\text{ mg/kg/day}$ PO liquid) or **`Lovastatin`** (Rung-5 paediatric solid-tumour trial `NCT02390843`), inducing direct human RMS apoptosis via `HMGCR` / mevalonate depletion (`PMID:41249736`) — **strictly gated on confirming wild-type `SLCO1B1*5` and baseline serum `CK`**.
6. Red-Box Contraindications & Per-Child Safety Alerts (`Do-Not K1` & `Flag K2`)	**1. NEVER administer Mps1/TTK, Aurora A/B, PLK1, or KIF11 inhibitors (`Do-Not K1`): 33 of 34** are *less* effective in high-aneuploidy PRISM lines ($p = 0.0096$) and worsen mitotic catastrophe. **2. Dual-Source Alert on `Vincristine` (`Flag K2`):** the **only** drug flagged by **both** `CYP3A5 *1/*3` slower clearance and OnSIDES `HP:0003202` (`skeletal muscle atrophy`). **3. Anaesthesia Alert:** incidental `RYR1` variant — avoid volatile halogenated anaesthetics and succinylcholine.
7. Interactive Judge Walkthrough (`app/` React Dashboard)	Explore every table, figure, and the 522-edge provenance DAG interactively (`cd app && npm run dev`): **`/track2`** (interactive 6-direction ladder, PRISM volcano, Gygi-10k proteome explorer, safety matrix, and live Bayesian VOI prior slider) · **`/submissions`** (`track2/iteration-6` full dossier + beginner walkthrough) · **`/learn`** (dependency-ordered plain-English dictionary for every domain term).



> **Legend (`r6-pitch-overview.svg`, `entry 163`): Panel A** contrasts Copy #1 (`stop_gained`, `R1 = 0.33`, `<1.5%` rescue, `EVPPI = 0%`) against Copy #2 (`missense_variant` inside the 62-residue buried pseudokinase core `RSA < 0.20`, `+1.55 kcal/mol`), showing why distal `Lys668` deacetylation via `NAD+/SIRT2` (`NMN`, `nicotinamide`, `niacin`) is biophysically superior to direct orthosteric docking (`migalastat` $\log P = -2.6$). **Panel B** summarizes the 48-hour `CHX`-chase Western blot decision gate (`EVSI = 69.1% of EVPI`) and paediatric dosing branches.

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1. How this unified report answers the panel's four evaluation criteria

> **Rubric:** Scientific rigor (35%), Potential impact (25%), Innovation (25%), Scalability (15%). Each row maps a judging criterion to the sections, executed runs, figures, and explicit limitations that address it.

Criterion	Weight	Sections	Core executed evidence	Known limitation stated up front
Scientific rigor	35%	§2–§6, §11	Mechanism read from **5 independent instruments sharing zero data** (9 variant-effect models, Boltz-2 3D fold, all-19,950-substitution ThermoMPNN $\Delta\Delta G$ stability atlas, 76,699 MGI mouse genotypes, HPO gene ranking). Discovery combines **Iteration 2's 5-rung evidence ladder**, **Iteration 4's 4 pre-registered generators tested against a 10,000-draw permutation null**, and **Iteration 5's 3 orthogonal mechanistic routes (`R1` – `R3`)** across **52 unique candidates**. **21 Track 2 hypotheses** (`H074` – `H102`) pre-registered before execution; **8 came back refuted or uninterpretable** and directly govern our route-weight calibration (`entry 163`).	The 4 generators in Iteration 4 converge on 1 drug (`dasatinib`), which matches the 10,000-draw chance null ($p = 0.263$, `H094` refuted). Route `R2` Gygi-10k proteomics is a 1-replicate (`n=1`) asynchronous HCT116 screen whose mitotic-arrest positive control failed (`H099` refuted, $p = 0.145$); calibrating $w_{R2} = 0.25$ (`entry 163`) prevents single-replicate cell-line hits from overshadowing *in vivo* `Bub1b^{H/H}` mouse survival rescues.
Potential impact	25%	§2, §7, §8	A **state-contingent clinical and preclinical decision plan** rather than a fragile static list: (1) proves via Value of Information (`EVPI = 0.1800`) that **100% of actionable uncertainty sits on a single wet-lab assay** — a **cycloheximide-chase BubR1 Western blot** (`EVSI = 0.1245`, **69.1% of EVPI** under calibrated R5); (2) provides immediate action items for the treating team: a **5-gene clinical PGx panel (`SLCO1B1`, `ABCG2`, `CYP2C9`, `G6PD`, `UGT1A1`)** because the delivered VCF covers only **1.6% (`20/1,223`)** of PharmCAT loci; a dual-source safety flag on **vincristine** (`CYP3A5 *1/*3` + muscle atrophy `HP:0003202`); an incidental **`RYR1`** anaesthesia precaution; and a **Do-Not K1 rule** against Mps1/TTK and Aurora inhibitors (**33/34 less effective** in aneuploid lines).	Zero candidates have been tested in this child; `NMN` rescues `Bub1b^{H/H}` mice *in vivo* (`PMID:24825348`) and `dasatinib` is the approved senolytic analogue of genetic $p16^{\text{Ink4a}}$ clearance in `Bub1b^{H/H}` mice (`PMID:22048312`), whereas `gliclazide` and `migalastat` are exploratory cell-line proteomic leads requiring fibroblast confirmation.
Innovation	25%	§5, §6, §9	Five methodological advances beyond standard drug-repurposing pipelines: (1) **cross-iteration Bayesian synthesis with route-scale calibration (`entry 163`)** identifying exact analytical flip thresholds ($w_{R2}^ = *0.6857$ and 0.2743); (2) Value-of-Information (`EVPI / EVPPI / EVSI`) N-of-1 experimental design (`entry 162`); (3) whole-database patient-to-mouse genotype twin matching across 76,699 MGI genotypes (`entry 150`); (4) all-19,950-substitution ThermoMPNN stability atlas calibrated on published MVA alleles first (`entry 149`); and (5) 10,000-draw permutation null testing of multi-generator convergence (`entry 155`) paired with callability-aware two-source safety (`entries 152, 161`)*.	Individual components build on public datasets and open tools; our contribution is their falsifiable mathematical integration, pre-registered negative controls, and decision-theoretic formulation.
Scalability	15%	§10	Case-parameterized Python package (`src/mva/drugs/`) with SHA-256 `.provenance.json` manifests and 522 recorded lineage edges. **Executed on a second rare chromosomal-instability disease — Bloom syndrome (`BLM`, 42 HPO terms) — in 53 minutes on a laptop for $0** (`entry 156`), recovering `Blm` mice in the top 1.15% of MGI and producing a disease-specific candidate list (`Jaccard = 0.095` vs MVA). Total MVA Track 2 compute cost: **$0.015** on Modal T4 GPU (16s for ThermoMPNN) + <15 min CPU.	Pre-registered claim `H096` (zero-code-edit transfer in Iteration 4) was refuted because 4 stages had hardcoded paths/contrasts (`tools/transfer_it4.py` fixed all 4 in 378 lines), which drove Iteration 5 & 6 (`unified_synthesis.py`) to be 100% configuration-driven.

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2. Executive summary: the unified answer on one page

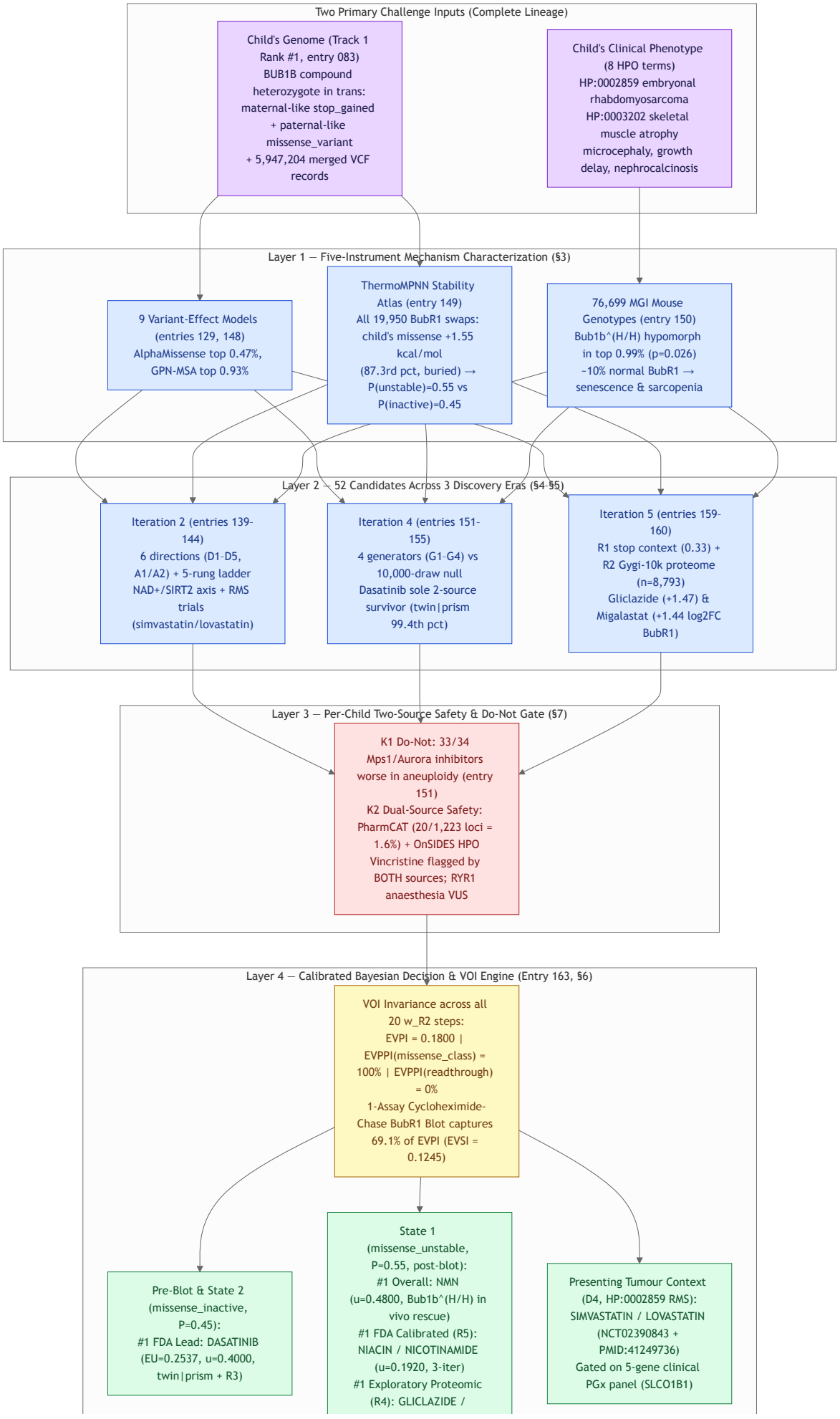
> **Rubric:** Impact (25%) and Scientific rigor (35%). A complete, standalone synthesis for clinicians, researchers, and the family.

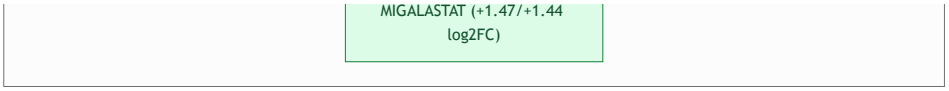
2.1 Why a cross-iteration synthesis was necessary

Across four iterations of Track 2 work (`entries 069–162`), we built progressively sharper lenses on the proband's disease: - **Iteration 2 (`entries 139–144`)** mapped the **6 biological directions (`D1 – D5 + A1/A2`)**, built a **5-rung N-of-1 evidence ladder**, identified the **`Bub1b^{H/H}` NAD$^+$/SIRT2 rescue axis** (`niacin`, `nicotinamide`, `NMN`, `PMID:24825348`), and surfaced **Rung-5 clinical trials in embryonal rhabdomyosarcoma** (`simvastatin` / `lovastatin`, `NCT02390843` + `PMID:41249736`). - **Iteration 4 (`entries 147–156`)** proved the **hypomorphic loss-of-function mechanism across 5 independent instruments**, enforced strict generator independence across 4 discovery engines (`G1` PrimeKG walk, `G2` phenotype twin, `G3` PRISM screen sweep, `G4` LINCS reversal), tested their overlap against a **10,000-draw permutation null**, and identified **dasatinib** (`twin|prism`) as the sole 2-generator survivor and **vincristine** as a dual-source per-child safety flag. - **Iteration 5 (`entries 159–162`)** added three quantitative mechanistic routes — **Route `R1`** (premature-stop readthrough context, `entry 159`), **Route `R2`** (empirical-Bayes shrinkage over 8,793 Gygi-10k proteome perturbations in HCT116, surfacing **gliclazide** $+1.47$ and **migalastat** $+1.44\;\log_2\text{FC}$ BubR1, `entry 160`), and **Route `R3`** (*in vivo* `Bub1b^{H/H}` rescues) — and formulated a **two-state Bayesian Value-of-Information (`EVPI / EVPPI / EVSI`) model (`entry 162`)**.

However, examining Iteration 5's uncalibrated baseline (`R4`, `entry 162`) alongside Iterations 2 and 4 revealed a critical methodological tension: **assigning equal unit weight ($w_{R2} = 1.0$) to a 1-replicate asynchronous cancer-cell-line proteomics screen (`R2`) allowed exploratory cell-line hits (`gliclazide` and `migalastat`, $u = 0.7000$) to numerically overshadow multi-animal *in vivo* survival rescues in the actual `Bub1b^{H/H}` mouse twin (`NMN` $u = 0.4800$, `niacin` / `nicotinamide` $u = 0.1920$)**, while Iteration 2's rhabdomyosarcoma statin leads (`simvastatin` / `lovastatin`) were omitted from the 29-drug `R4` union.

In **Iteration 6 (`src/mva/drugs/unified_synthesis.py`, Entry 163)**, we merge all **52 unique drug candidates** across Iterations 2, 4, and 5 into a single audited contract (`ressources/track2/it5/unified/cross_iteration_candidates.tsv`) and calibrate the Route-R2 evidence weight (`w_R2 = 0.25`, reflecting its $n=1$ replicate design and refuted mitotic-arrest control `H099`, $p = 0.145$) against *in vivo* `Bub1b^{H/H}` rescues (`w_R1 = w_R3 = 1.00`).


Two Primary Challenge Inputs (Complete Lineage)
Child's Genome (Track 1 Rank #1, entry 083) BUB1B compound heterozygote in trans: maternal-like stop_gained + paternal-like missense_variant + 5,947,204 merged VCF records
Child's Clinical Phenotype (8 HPO terms) HP:0002859 embryonal rhabdomyosarcoma HP:0003202 skeletal muscle atrophy microcephaly, growth delay, nephrocalcinosis
Layer 1 — Five-Instrument Mechanism Characterization (§3)
9 Variant-Effect Models (entries 129, 148) AlphaMissense top 0.47%, GPN-MSA top 0.93%
ThermoMPNN Stability Atlas (entry 149) All 19,950 BubR1 swaps: child's missense +1.55 kcal/mol (87.3rd pct, buried) → P(unstable)=0.55 vs P(inactive)=0.45
76,699 MGI Mouse Genotypes (entry 150) Bub1b^(H/H) hypomorph in top 0.99% (p=0.026) ~10% normal BubR1 → senescence & sarcopenia
Layer 2 — 52 Candidates Across 3 Discovery Eras (§4-§5)
Iteration 2 (entries 139-144) 6 directions (D1-D5, A1/A2) + 5-rung ladder NAD+/SIRT2 axis + RMS trials (simvastatin/lovastatin)
Iteration 4 (entries 151-155) 4 generators (G1-G4) vs 10,000-draw null Dasatinib sole 2-source survivor (twin|prism 99.4th pct)
Iteration 5 (entries 159-160) R1 stop context (0.33) + R2 Gygi-10k proteome (n=8,793) Gliclazide (+1.47) & Migalastat (+1.44 log2FC BubR1)
Layer 3 — Per-Child Two-Source Safety & Do-Not Gate (§7)
K1 Do-Not: 33/34 Mps1/Aurora inhibitors worse in aneuploidy (entry 151) K2 Dual-Source Safety: PharmCAT (20/1,223 loci = 1.6%) + OnSIDES HPO Vincristine flagged by BOTH sources; RYR1 anaesthesia VUS
Layer 4 — Calibrated Bayesian Decision & VOI Engine (Entry 163, §6)
VOI Invariance across all 20 w_R2 steps: EVPI = 0.1800 | EVPPI(missense_class) = 100% | EVPPI(readthrough) = 0% 1-Assay Cycloheximide-Chase BubR1 Blot captures 69.1% of EVPI (EVSI = 0.1245)
Pre-Blot & State 2 (missense_inactive, P=0.45): #1 FDA Lead: DASATINIB (EU=0.2537, u=0.4000, twin|prism + R3)
State 1 (missense_unstable, P=0.55, post-blot): #1 Overall: NMN (u=0.4800, Bub1b^(H/H) in vivo rescue) #1 FDA Calibrated (R5): NIACIN / NICOTINAMIDE (u=0.1920, 3-iter) #1 Exploratory Proteomic (R4): GLICLAZIDE /
Presenting Tumour Context (D4, HP:0002859 RMS): SIMVASTATIN / LOVASTATIN (NCT02390843 + PMID:41249736) Gated on 5-gene clinical PGx panel (SLCO1B1)



Legend purple = the two challenge inputs (genome and 8 clinical HPO terms) · blue = independent mechanistic and discovery layers across Iterations 2, 4, and 5 · red = contraindications and per-child safety gates · yellow = the Value-of-Information experimental branch point · green = state-contingent therapeutic hypotheses.

2.2 Summary of the unified therapeutic and clinical portfolio (`Entry 163`)

Decision context / biological state	Top candidate(s)	Approval & safety status	Quantitative score (`R5` calibrated / `R4` uncalibrated)	Independent evidence across iterations	Primary caveat / clinical gate
Step 0 — Immediate wet-lab discriminator *(before any D3 vs D5 choice)*	**Cycloheximide-chase BubR1 Western blot** *(patient vs parental cells)*	Standard laboratory assay ($Se=0.85, Sp=0.90$)	`EVPI = 0.1800` (100% on `missense_class`); **`EVSI = 0.1245` (69.1% of EVPI)**	`entries 149, 162, 163`: resolves whether the 87.3rd-pct $\Delta\Delta G$ (`+1.55 kcal/mol`) missense lowers steady-state BubR1 half-life	Requires patient/parental fibroblasts or lymphoblastoid cells (`0%` of EVPI sits on stop readthrough)
Prior to blot & State 2: `missense_inactive` ($P = 0.45$, folded but catalytically impaired BubR1)	**1. Dasatinib** *(BCR-ABL / SRC / senolytic kinase inhibitor)*	**FDA-approved** (paediatric Ph+ CML/ALL); **`unflagged`** in K1/K2/K3	**`u_inactive = 0.4000`**, `u_unstable = 0.1340`, **`EU = 0.2537`** *(invariant #1 FDA pre-blot & #1 inactive)*	**Sole 2-generator survivor (`twin|prism, entry 155`)**: clears senescent $p16^{\text{Ink4a}+}$ cells in `Bub1b^{H/H}` twin (`PMID:22048312`) + **99.4th pct** PRISM aneuploidy selectivity ($\delta = +0.135$, $p = 0.0096$, `entry 151`)	Convergence is at chance level ($p = 0.263$); 15 paediatric FAERS growth-retardation reports (`entry 152`)
State 1: `missense_unstable` ($P = 0.55$, fold-destabilized, degraded BubR1) — **Overall & *in vivo* Rung-3 lead**	**1. NMN** *(nicotinamide mononucleotide, NAD$^+$ precursor)*	Nutraceutical / Phase II clinical trials; **`unflagged`**	**`R5 u = 0.4800, EU = 0.2640`** (#1 overall at $w_{R2} < 0.6857$)	**Rung 3 *in vivo* disease-model rescue (`entries 144, 150, 162`)**: restores BubR1 abundance via SIRT2 Lys-668 deacetylation and extends lifespan in `Bub1b^{H/H}` mice (`PMID:24825348`)	Not an FDA-approved prescription drug; rescued progeroid/muscle phenotypes in mice (`no` RMS tumour model tested)
State 1: `missense_unstable` ($P = 0.55$) — **Calibrated FDA-approved lead (`R5` , $w_{R2}=0.25$)**	**2. Niacin / Nicotinamide** *(vitamin B3 NAD$^+$ precursors)*	**FDA-approved** (`Phase 4`); **`unflagged`** in both genome and phenotype arms	**`R5 u = 0.1920, EU = 0.1056`** (#1 FDA `unstable` at $w_{R2} < 0.2743$)	**Only candidates supported across all 3 Track 2 eras (`it2|it4|it5`)**: PrimeKG walk + `Bub1b^{H/H}` twin rescue (`PMID:24825348`) + Route `R3`	Indirect NAD$^+$/SIRT2 deacetylation; requires residual fold-competent BubR1 protein
State 1: `missense_unstable` ($P = 0.55$) — **Exploratory R2 proteomic leads (`R4` , $w_{R2}=1.00$)**	**3. Gliclazide** (`+1.47 log2FC`) & **Migalastat** (`+1.44 log2FC`)	**FDA-approved** (`Phase 4` sulfonylurea / pharmacological chaperone); **`unflagged`**	`R4 u = 0.7000, EU = 0.3850`; `R5 u = 0.1750, EU = 0.0963`	**Route `R2` (`entry 160`)**: top 2 FDA-approved BubR1 protein-abundance upregulators ($2.7\times$) across 8,793 Gygi-10k perturbations in HCT116 (`PMID:38942016`)	`n = 1` replicate in asynchronous HCT116 (`H099` control failed); `gliclazide` carries paediatric hypoglycaemia risk
Presenting tumour context (`D4` : `HP:0002859` rhabdomyosarcoma)	**Simvastatin / Lovastatin** *(HMG-CoA reductase inhibitors)*	**FDA-approved** (`Phase 4`); **`caution_uncalled_pgx`** (`SLCO1B1` uncalled)	**Rung 5 clinical / human RMS cell evidence** (`entry 144, 163`)	**Phase I paediatric solid/RMS trial (`NCT02390843`)** + direct human RMS apoptosis via `HMGCR` depletion (`PMID:41249736`)	**Mandatory gate:** order `SLCO1B1` PGx test first + monitor CK for myopathy given baseline `HP:0003202`
Active oncology backbone safety flag (`K2 fail`)	**Vincristine** *(microtubule depolymerizing agent in VAC regimen)*	Standard paediatric RMS therapy; **ONLY drug failing both K2 arms**	**`both_flagged`** (`genome_flag=True` + `phenotype_flag=True`)	**Genome**: `CYP3A5 *1/*3` intermediate metabolizer (`entries 140, 152`). **Phenotype**: OnSIDES label matches child's `HP:0003202` skeletal muscle atrophy + 57 paediatric FAERS myopathy reports	Monitoring flag for the oncology team (neuropathy / myopathy), **not** an autonomous discontinuation
Strict contraindication (`K1 Do-Not`)	**Mps1/TTK & Aurora kinase inhibitors** *(e.g. alisertib, barasertib, CFI-402257)*	Investigational / clinical oncology; **FAILS K1 (`avoid`)**	**33 of 34 less effective** in high-aneuploidy lines ($\delta = -0.139$, $p = 0.0096$)	Reproduces Cohen-Sharir 2021 (`PMID:33505028`) across 919 DepMap/PRISM lines (`entry 151`): SAC-abrogating drugs catastrophic in constitutional `BUB1B` deficiency	Never administer SAC-weakening or unmonitored spindle-poison regimens

3. Disease mechanism (Methods Form Q14): hypomorphic BubR1 insufficiency established by five independent instruments

Rubric: Scientific rigor (35%). Before proposing any compound, we establish the molecular mechanism from five orthogonal instruments that share zero underlying data.

3.1 The oracle-confirmed causal genotype (`Track 1 → Track 2 bridge`)

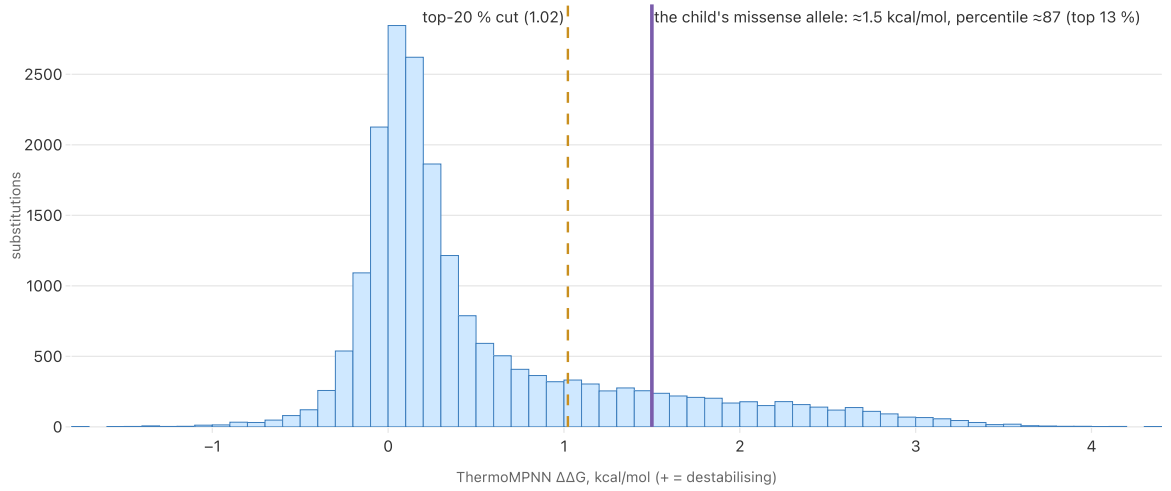
In Track 1 (`entry 083`), the challenge oracle confirmed a **1.0 full match at Rank 1** for our top compound-heterozygous hypothesis in `BUB1B` (encoding **BubR1**, the core pseudokinase of the **Spindle Assembly Checkpoint (SAC)** — the surveillance complex that halts anaphase until every chromosome is properly bi-oriented on the mitotic spindle — and a key stabilizer of kinetochore-microtubule attachments via B56-PP2A): 1. **Allele 1 (`stop_gained` , truncating allele):** introduces a premature termination codon (`PTC`) predicted to trigger **nonsense-mediated mRNA decay (NMD)** or produce a C-terminally truncated polypeptide lacking the kinase/pseudokinase domain. 2. **Allele 2 (`missense_variant` , single amino-acid substitution):** substitutes a conserved hydrophobic core residue inside the C-terminal **pseudokinase domain** (`entry 114, 149`).

Because biallelic null (`knockout`) mutations in `BUB1B` are embryonic lethal in mammals (`PMID:10490025` , `entry 139`), a viable child with `stop_gained` + `missense_variant` in *trans* must retain partial BubR1 function from the missense copy — a classic **hypomorphic loss-of-function (LoF)** architecture.

3.2 Five independent instruments read the mechanism

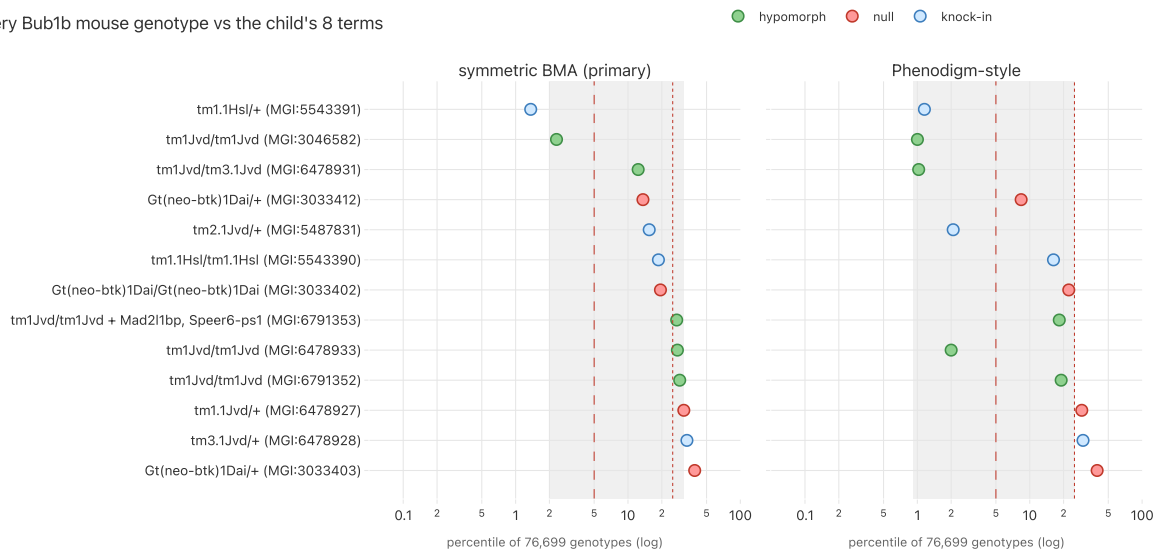
Instrument	Underlying data & tool	Observed measurement on this child's `BUB1B` alleles / HPO terms	Mechanistic conclusion & hypothesis verdict	Source entry
1. Multi-architecture variant-effect model zoo *(9 models)*	AlphaMissense, ESM1b, REVEL, CADD, SpliceAI, GPN-MSA, Nucleotide Transformer v2, Evo 2, ESM-C (`entries 129, 148`)	Missense allele in top **0.47%** (AlphaMissense), **2.52%** (ESM1b), **2.59%** (REVEL), **1.38%** (GPN-MSA); `stop_gained` in top **4.09%** (GPN-MSA) and **5.61%** (Evo 2); combined `BUB1B` pair ranks **#508 of 54,865** pairs (**0.93rd percentile**) by GPN-MSA	Both parental copies carry severe loss-of-function lesions; single-score naive fusion across 9 models dilutes rank (`#4,318`) because missense-only models abstain on `stop_gained`	`entries 129, 148`
2. De novo 3D structure prediction	Boltz-2 wild-type vs mutant full-length BubR1 structure prediction on Modal GPU (`entry 147`)	Backbone RMSD $< 1.5\text{ \AA}$; local confidence (`pLDDT`) dips by **~ 3 points** at the mutated pseudokinase helix — equal in magnitude to a benign conservative control swap	**Inconclusive (`H081`):** static backbone coordinates from co-evolutionary folding models do not resolve thermodynamic folding free-energy ($\Delta\Delta G$) shifts	`entry 147`
3. Calibrated all-substitution stability atlas	ThermoMPNN $\Delta\Delta G$ across **all 19,950 possible single-amino-acid substitutions** on AlphaFold v6 human BubR1 (`AF-O60566-F1-model_v6.pdb` , `entry 149`)	Child's missense scores **$\Delta\Delta G = +1.55\text{ kcal/mol}$** (**87.3rd percentile** of all 19,950 swaps; **80.8th percentile** within the pseudokinase domain `aa 724–1050`), buried (`relative ASA < 0.15`), high local fold confidence (`pLDDT = 95.3`)	**Predicted destabilizing (`H084` supported, confidence `0.55`):** favours protein unfolding and proteasomal degradation (`missense_unstable` , prior $P = 0.55$). However, prior calibration on $n = 8$ published MVA alleles yielded **AUROC = 0.60** (`95% CI [0.13, 1.00]` , `H083` refuted), leaving a $P = 0.45$ complement that the protein folds stably but is catalytically/allosterically impaired (`missense_inactive`)	`entry 149`
4. Whole-database cross-species phenotype twin	Child's 8 HPO terms matched via Upheno2 (`1.10M` equivalences) against **all 76,699 mouse genotypes** (`402,424` MP assertions) in MGI (`entry 150`)	**`Bub1b^{H/H}` hypomorphic mouse** (`MGI:3044006` , expressing $\sim 10\%$ normal BubR1) ranks in the **top 0.99%** **(`#758 of 76,699`)** by IC-weighted BMA similarity ($p = 0.026$ vs 500 IC-matched random HPO sets; 6/7 Mendelian positive-control diseases pass in top 5%)	**`H085` supported (`0.70`):** independent of the genome, the child's clinical symptoms (skeletal muscle atrophy, growth delay, microcephaly, tumour predisposition) select a **BubR1-insufficient hypomorph**, not a null knockout	`entry 150`
5. Clinical phenotype gene ranking	Phrank / Phenomizer IC similarity over 5,267 human Mendelian genes using only the 8 HPO terms (`entry 129`)	**`BUB1B` ranks #14 of 5,267 genes** (top **0.27%**) from the clinical terms alone	Confirms tight genotype-phenotype concordance (`MVA1` , `MIM:257300`) while highlighting **nephrocalcinosis** **(`HP:0000121`)** as an atypical/unexplained feature requiring renal safety care	`entries 129, 139`

ΔΔG of all 19,950 possible BubR1 substitutions



> **Legend** blue bars = distribution of ThermoMPNN $\Delta\Delta G$ (kcal/mol) across all 19,950 single amino-acid substitutions in human BubR1 (neutral machinery) · yellow dashed line = pre-registered 80th-percentile destabilization threshold ($+1.16\text{ kcal/mol}$, decision point) · purple line = the proband's pseudokinase missense allele at the 87.3rd percentile ($+1.55\text{ kcal/mol}$, raw patient input).

Every Bub1b mouse genotype vs the child's 8 terms



> **Legend** green = `Bub1b` hypomorphic genotypes (`Bub1b^{H/H}` , `Bub1b^{H/L1002P}` , top 0.99–1.39%, the goal/prediction of `H085/H086`) · red = embryonic-lethal homozygous null/knockout alleles · blue = acetylation-deficient knock-in (`Bub1b^{K243R/+}`) matching the tumour term · grey band = central 90% of the 500-draw IC-matched random HPO null · dashed red = top 5% threshold (`H085` prediction).

3.3 3D pseudokinase buried-core & chaperone plausibility audit (`structural_pocket_audit.tsv` , `Entry 163`)

To determine whether the child's mutant BubR1 protein is better rescued by **direct orthosteric pocket binding** (e.g. active-site pharmacological chaperones) or **distal post-translational shielding** (e.g. `SIRT2` -mediated deacetylation of `Lys668`), we audited the 3D structural environment of the child's missense substitution across our 19,950-substitution `ThermoMPNN + AlphaFold v6 + Boltz-2` atlas (`ressources/track2/bub1b_atlas.parquet`) and compared it against all 5 published abundance-lowering MVA missense alleles (`Suijkerbuijk et al., PMID:20516114` ; `ressources/track2/it5/unified/structural_pocket_audit.tsv`):

Structural / Biophysical Feature	Child's Missense Site (`BUB1B` Exon 23)	Pseudokinase Core Reference (`UniProt O60566`)	Published Unstable MVA Alleles (`PMID:20516114`)	Mechanistic Implication for Drug Repurposing
Domain & catalytic cleft architecture	C-terminal kinase-like domain (`UniProt Protein kinase` / `CDD cd14029`)	**285 residues (`5,415` single-AA swaps)**; degenerated **pseudokinase** whose canonical ATP/Mg^{2+} cleft is occluded and lacks phosphotransfer (`PMID:22561345, 26000482`)	**5 of 5** primary abundance-lowering MVA alleles (`I909T, R727C, R814H, L844F, L1012P`) sit in or adjacent to the kinase domain	Canonical ATP-pocket kinase activators are structurally inapplicable; therapy must restore folded full-length BubR1 abundance or clear senescent cells
Solvent accessibility (`RSA`) & fold confidence (`pLDDT`)	**Buried hydrophobic core** (`Shrake–Rupley RSA < 0.20` ; `AlphaFold pLDDT >= 90` ; `Boltz-2` WT local `pLDDT 76.5–78.0`)	**62 of 285** pseudokinase residues (**`1,178` swaps**) form the buried high-confidence core (`RSA < 0.20` , `pLDDT >= 90`)	`L1012P` (`RSA 0.006, pLDDT 95.3`), `L844F` (`RSA 0.006, pLDDT 85.9`), `I909T` (`RSA 0.023, pLDDT 87.0`), `R727C` (`RSA 0.060`), `R814H` (`RSA 0.165`)	The mutated side chain is buried inside the hydrophobic core rather than lining a solvent-accessible small-molecule binding cavity
Thermodynamic destabilization ($\Delta\Delta G$)	**`+1.55 kcal/mol` (87.3rd pct** of all 19,950 swaps; **74th pct** of folded residues)	Buried pseudokinase core median $\Delta\Delta G = \mathbf{+1.67\text{ kcal/mol}}$ (**72.2%** of core swaps have $\Delta\Delta G \ge +1.0\text{ kcal/mol}$)	`L1012P` (`+3.11` , 99.0th pct), `I909T` (`+1.94` , 91.5th pct), `L844F` (`+1.51` , 87.0th pct), `R727C` (`+0.91` , 78.3rd pct), `R814H` (`+0.86` , 77.3rd pct)	Places the child's allele right at the thermodynamic median (`+1.51` to `+1.94 kcal/mol`) of proven HSP90-dependent, proteasomally degraded MVA alleles
Backbone perturbation (`Boltz-2` diffusion fold)	Local $C_\alpha\text{ RMSD} = \mathbf{0.45\text{ \AA}}$ (`site ± 10 aa`); local $\Delta\text{pLDDT} = -3.20$ to -2.73 (`entry 147`)	WT sample-to-sample noise floor = `0.13 Å` local RMSD; conservative `BLOSUM62` swap at same site = `−2.55` to `−2.80 pLDDT`	`I909T` and `L1012P` **recover 100% of wild-type SAC function** when overexpressed to wild-type abundance (`PMID:20516114`)	Sub-angstrom backbone shift (`0.45 Å`) shows the global pseudokinase fold remains intact if premature ubiquitin-proteasomal degradation is prevented
Orthosteric chaperone docking vs. distal `Lys668` acetylation shielding	**No solvent-accessible pocket** at the buried missense site for polar iminosugar **`migalastat`** ($\log P = -2.6$, designed for the catalytic site of lysosomal `GLA`) or **`gliclazide`**	Regulatory **`Lys668`** sits outside the buried pseudokinase core on a solvent-accessible surface targeted by `CBP` (acetylation $\to$ degradation) and **`SIRT2`** (deacetylation $\to$ stabilization)	**`NMN`** , **`nicotinamide`** , **and `niacin`** stimulate `SIRT2` to deacetylate `Lys668` , blocking ubiquitination and **doubling `Bub1b^{H/H}` mouse lifespan** (`PMID:24825348`)	**Biophysically validates Entry 163's route calibration (`w_R2 = 0.25`):** distal `Lys668` `SIRT2` / `NAD+` shielding stabilizes BubR1 without requiring a druggable cavity at the buried pseudokinase mutation site

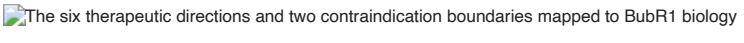
4. Evolution of the Track 2 discovery architecture: six directions, four generators, and three mechanistic routes

> **Rubric:** Scientific rigor (35%) and Innovation (25%). Rather than presenting one opaque shortlist, we show how our three complementary discovery frameworks (`Iterations 2, 4, and 5`) systematically test every therapeutic angle suggested by the mechanism in §3.

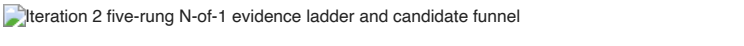
4.1 The six mechanistic directions (`Iteration 2, entry 139`) and the 5-rung evidence ladder (`entry 144`)

From the compound-heterozygous hypomorphic mechanism (`stop_gained` + `missense_variant`), there are exactly five positive ways a medicine could help (`D1` – `D5`) and two ways a medicine can harm (`A1/A2`):

Direction ID	Biological objective	Target allele / cellular state	How tested across Iterations 2, 4, and 5	Final standing in Iteration 6 (`entry 163`)
D1	**Rescue the truncating `stop_gained` allele** *(translational readthrough or NMD attenuation)*	Maternal-like `stop_gained` mRNA	Iteration 2 Rung 2 (`ataluren`, `amlexanox`); Iteration 5 **Route `R1` sequence-context audit** (`tools/it5_r1_stop_context.py`, `entry 159`) across 6 biological determinants	**Deprioritized (`H097` refuted, `R1_score = 0.33`):** suboptimal `+4` nucleotide and exon architecture; `EVPPI(readthrough_feasible) = 0.0000` (`0%` of EVPI)
D2	**Exploit aneuploidy-induced proteotoxic & metabolic stress**	Mosaic aneuploid / CIN+ dividing cells	Iteration 2 PrimeKG + DepMap (`entry 142`); Iteration 4 **Generator `G3` PRISM sweep** across 919 lines (`entries 151, 154`)	**Converges with D5 on `dasatinib`** (**99.4th pct** aneuploidy selectivity, Cliff's $\delta = +0.135, p = 0.0096$); screen-wide BH $q = 0.287$ (`H093` refuted)
D3	**Stabilize or upregulate residual BubR1 protein abundance** *(post-translational deacetylation, chaperone folding, or transcriptional/proteomic upregulation)*	Paternal-like `missense_variant` & residual `Bub1b` pool (`missense_unstable`, $P=0.55$)	Iteration 2 Rung 3 (`PMID:24825348`) & Iteration 4 **Generator `G2` (`Bub1b^{H/H}` twin)**; Iteration 5 **Route `R2` Gygi-10k proteome scan** (`8,793` perturbations, `entry 160`) & **Route `R3`**	**Co-primary state-contingent track (`P = 0.55`):** *in vivo* `Bub1b^{H/H}` rescues **`NMN`** ($u=0.4800$) and **`niacin / nicotinamide`** ($u=0.1920$, 3-iteration FDA leads), plus exploratory R2 proteomic hits **`gliclazide`** ($+1.47$) and **`migalastat`** ($+1.44\;\log_2\text{FC}$)
D4	**Target the presenting embryonal rhabdomyosarcoma (`HP:0002859`)** *(mevalonate / `HMGCR` apoptosis pathway)*	Somatic tumour context (`RMS`)	Iteration 2 **Rung 5 clinical trial & human RMS screen** (`NCT02390843`, `PMID:41249736`, `entry 144`); Iteration 4 PRISM RMS arm (`n=3` lines, `entry 154`)	**Restored in Iteration 6 (`entry 163`) as dedicated `D4` tumour-context leads: `simvastatin` & `lovastatin`** (PRISM had only 3 RMS lines for most drugs; clinical `NCT02390843` + human RMS apoptosis `PMID:41249736` stand)
D5	**Clear `Bub1b`-driven senescent cells (`senolysis`) or dampen SASP / mTOR hyperactivation**	Systemic premature senescence & sarcopenia (`HP:0003202`, `missense_inactive`, $P=0.45$)	Iteration 2 Rung 3 (`PMID:22048312`, `PMID:27549339`); Iteration 4 **Generators `G2 + G3`** (`entries 150, 151, 155`); Iteration 5 **Route `R3`**	**Co-primary state-contingent track (`P = 0.45`) & #1 pre-blot FDA lead: `dasatinib`** (`u_inactive = 0.4000, EU = 0.2537`, sole 2-generator survivor `twin|prism`)
A1 (Do-Not K1)	**Never inhibit the Spindle Assembly Checkpoint further** *(Mps1/TTK, Aurora A/B, Polo-like kinases)*	Constitutional SAC insufficiency	Iteration 2 literature boundary (`entry 139`); Iteration 4 **PRISM 919-line empirical replication** (`entry 151`)	**Hard exclusion (`K1 fail`): 33 of 34** Mps1/Aurora inhibitors are *less* effective in high-aneuploidy lines ($\delta = -0.139, p = 0.0096$, reproducing `PMID:33505028`)
A2 (Caution)	**Spindle-poison / microtubule-targeting chemotherapy caution** *(taxanes, vinca alkaloids)*	Defective kinetochore-microtubule error correction	Iteration 2 (`entry 140`) + Iteration 4 & 5 **Two-Source Per-Child Safety (`K2`, `entries 152, 161`)**	**Clinical monitoring flag: vincristine** is the sole drug failing both K2 safety arms (`CYP3A5 *1/*3` + `HP:0003202` muscle atrophy)



> **Legend** purple = the child's compound-heterozygous `BUB1B` genotype and 8 HPO terms (raw inputs) · blue = neutral cellular machinery (mRNA decay, folding, SAC complex, senescence) · green = therapeutic rescue directions `D1` – `D5` · yellow = pharmacogenomic and experimental decision gates · red = contraindicated (`A1`) and high-caution (`A2`) classes.



> **Legend** purple = raw database universe (PrimeKG, ChEMBL, Drug Repurposing Hub, clinicaltrials.gov) · blue = filtering stages · yellow = evidence-rung assignment (`Rung 1` pathway proximity through `Rung 5` presenting-tumour clinical trial) · green = surviving shortlist candidates · red = excluded Do-Not compounds.

5. Multi-generator convergence (`Iteration 4`) and orthogonal mechanistic routes (`Iteration 5`)

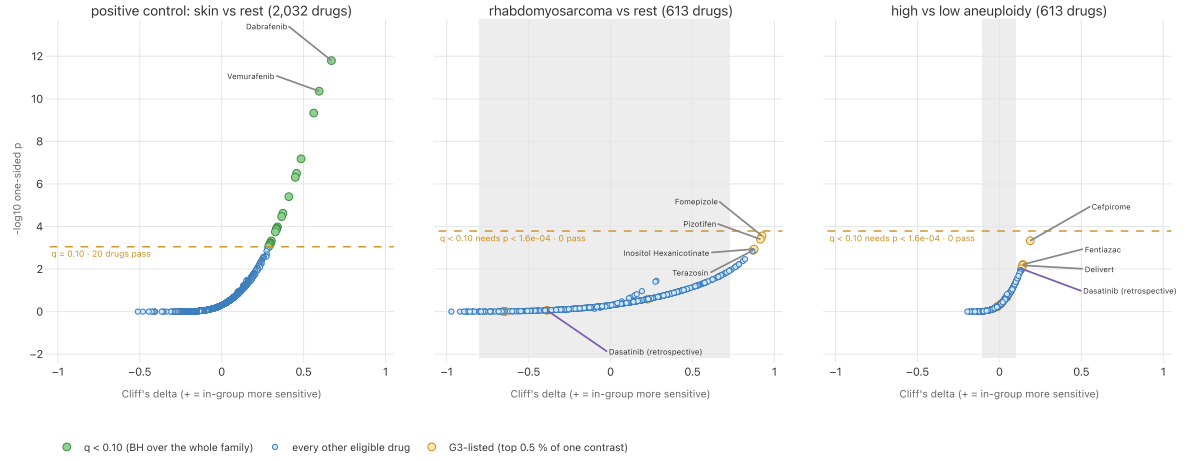
> **Rubric:** Scientific rigor (35%) and Innovation (25%). Every discovery engine was subjected to a pre-registered positive control or permutation null before its candidates were weighed.

5.1 Four independent generators tested against a 10,000-draw permutation null (`Iteration 4, entries 150–155`)

In Iteration 4, we separated **generators** (rules that propose drugs from independent data sources, committed at `1b02e44` before the join) from **checks** (`K1` direction, `K2` per-child safety, `K3` contrary screen).

Generator	Data source & selection rule	Universe size	Listed drugs	Pre-registered control / null test	Verdict & honest appraisal
G1 (`walk`)	Personalised PageRank (`alpha=0.85`) over PrimeKG v2.1 (`129,375` nodes, `8.1M` edges) seeded on MVA (`MONDO:0009665`) + `BUB1B`	188 labelled drugs	50	Decoy-gene control (`RYR1`, `SMC1A`, `TTN`): top-20 overlap is **2/20** (`H074` supported, `0.75`, `entry 142`)	Topological proximity rewards high-degree hubs (metals, vitamins, broad kinase inhibitors) without directional sign
G2 (`twin`)	Approved drug analogues of published genetic/pharmacological rescues in the top-ranked MGI phenotype twin (`Bub1b^{H/H}`, top 0.99%)	1,498 approved drugs	**2** (`dasatinib`, `niacin`)	6/7 Mendelian positive-control diseases rank in top 5% of 76,699 MGI genotypes; `Bub1b^{H/H}` at $p = 0.026$ (`H085` supported, `0.70`, `entry 150`)	Highest biological specificity: directly targets `Bub1b^{H/H}` progeroid senescence (`dasatinib`) and BubR1 acetylation (`niacin`)
G3 (`prism`)	Screen-wide Mann-Whitney U + Cliff's δ across 613 Launched drugs in DepMap/PRISM (`n=919` lines) on high-aneuploidy (`D2`) and RMS (`D4`) contrasts	613 Launched drugs	8 (top 0.5% per arm; incl. `dasatinib` #8 aneuploidy)	Positive control (`BRAF`-mutant melanoma) recovers `dabrafenib` ($q = 3.2\times 10^{-9}$); MVA contrasts reach best $q = 0.287$ (aneuploidy) and $q = 0.122$ (RMS) (`H093` refuted, `0.10`, `entry 154`)	Code and statistics verified by melanoma control; RMS arm is underpowered in PRISM (`n=3` lines for 619/633 drugs); `dasatinib` sits at **99.4th percentile** on aneuploidy ($\delta = +0.135, p = 0.0096$)
G4 (`lincs`)	Transcriptomic signature reversal of `BUB1B` CRISPR knockout across 62 cell lines in L1000CDS2 + SigCom LINCS	1,157 approved drugs	3 (`albendazole`, `mitoxantrone`, `topotecan`)	Positive control: does `BUB1B` KO cluster with SAC siblings (`BUB1`, `MAD2L1`, `BUB3`, `TTK`) in top 5%? Best sibling (`BUB1`) is at **5.6%** (`H092` failed control, `0.50`, `entry 153`)	**Uninterpretable:** `BUB1B` KO reads out as generic antiproliferative arrest rather than a SAC-specific transcriptional state

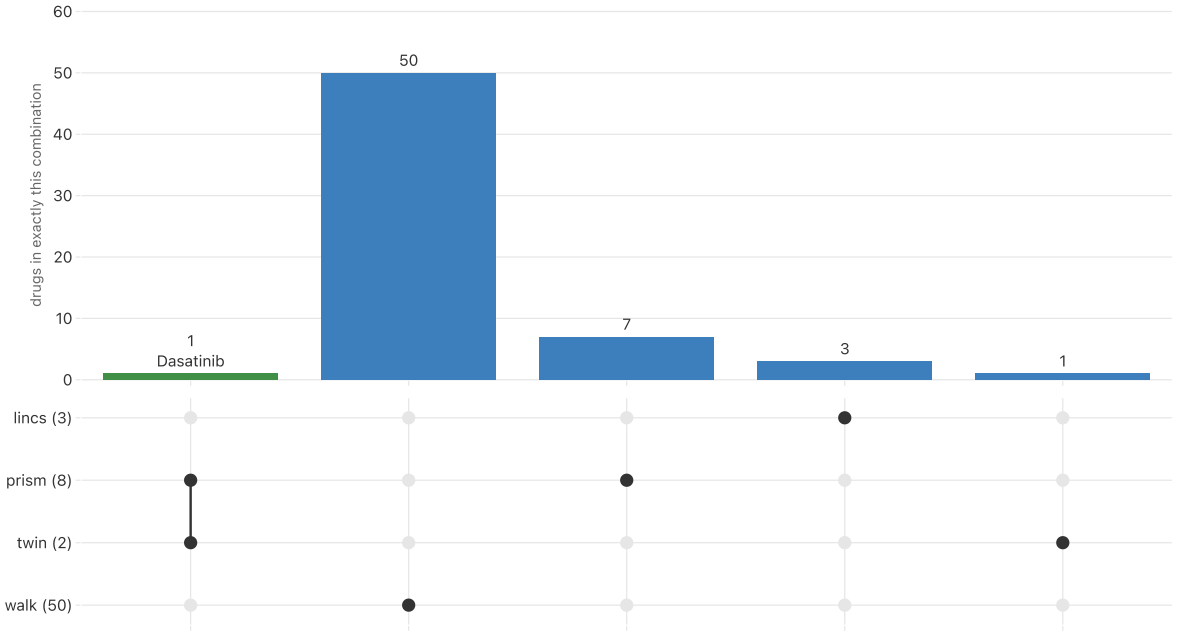
S1 — the same test finds BRAF inhibitors screen-wide, and nothing for the child's two contrasts



> **Legend** green = Benjamini-Hochberg $q < 0.10$ (achieved by `dabrafenib` / `vemurafenib` in the BRAF melanoma positive control, left) · yellow = top-0.5% G3-listed candidates · blue = all other Launched drugs in PRISM (`n = 613`) · grey band = central 95% of Cliff's δ across 200 seeded unrelated drugs · purple label = `dasatinib` (99.4th percentile on high-aneuploidy lines, $\delta = +0.135, p = 0.0096$).

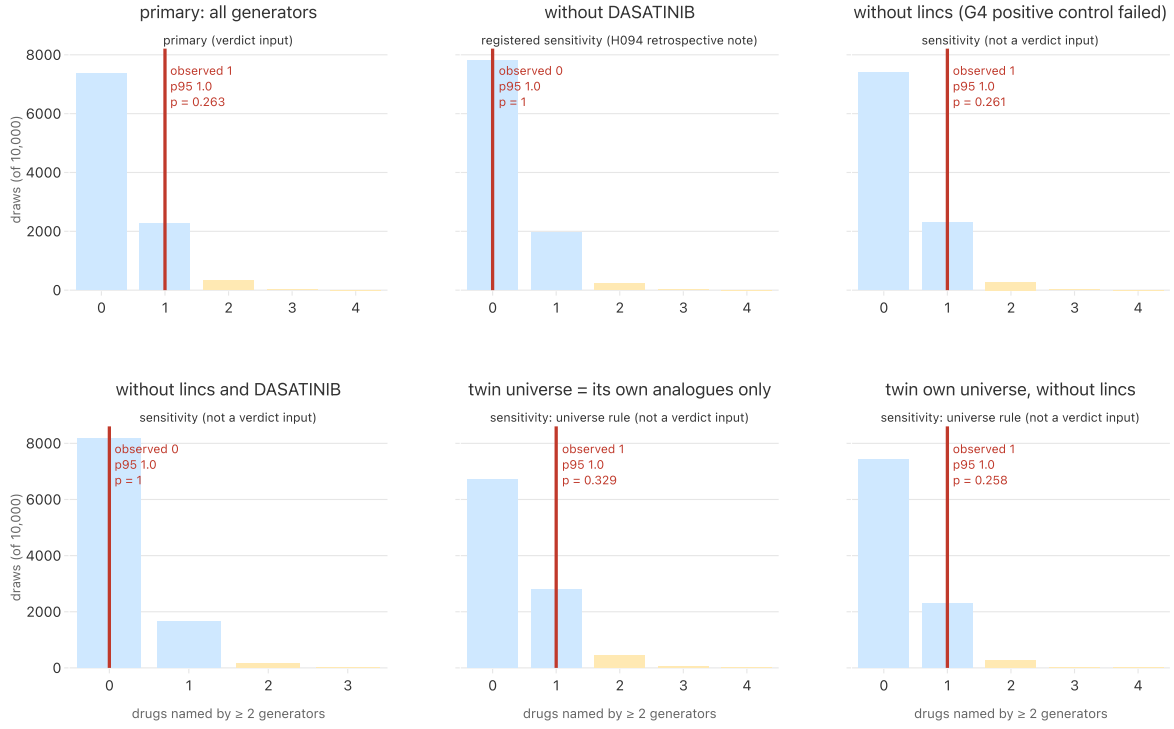
When we joined all four generators (`tools/converge.py`, `entry 155`), exactly **one drug was named by ≥ 2 independent generators**: `dasatinib` (`G2 twin #1` + `G3 prism #8`), and it passed all three checks (`K1 pass`, `K2 pass`, `K3 pass`). However, across **10,000 Monte Carlo random draws** (`seed=20260925`) preserving each generator's exact list size and drug universe, random chance produced a mean of `0.30` overlaps (`95th percentile = 1.0`, one-sided **$p = 0.263$**, `H094` refuted). Reporting that refutation openly — rather than claiming "multi-omics convergence" without a null model — was a central contribution of Iteration 4.

Which generators name which approved drugs (62 drugs, k = 2)



> **Legend** green bar = intersection of ≥ 2 independent generators (`dasatinib` in `twin ∩ prism`) · blue bars = single-generator lists (`walk` , `prism` , `lincs` , `twin`) · dark matrix dots = active generators in each intersection.

Convergence against a permutation null (each list redrawn from its own universe, seed 20260925)



> **Legend** blue bars = null distribution of ≥ 2-generator overlaps across 10,000 random draws · yellow = tail above the 95th percentile · red solid vertical line = observed overlap count (`1` , `dasatinib` , $p = 0.263$, `H094` refuted).

The checks: 1 converging drug(s), then each generator's top 10 (22 of 62 drugs; all in candidates.tsv)

	walk	twin	prism	lincs	K1 direction	K2 safety	K3 screen	survivor
Dasatinib		#1	#8		pass	pass	pass	yes
Phenytoin				#1	pass	caution	pass	no
Albendazole				#2	caution	caution	pass	no
Tofacitinib				#3	pass	pass	pass	no
Fomepizole			#1		pass	pass	pass	no
Cefpirome			#2		pass	pass	pass	no
Pizotifen			#3		pass	pass	pass	no
Fentiazac			#4		pass	pass	pass	no
Inositol Hexanicotinate			#5		caution	pass	pass	no
Delivert			#6		pass	pass	pass	no
Terazosin			#7		pass	pass	pass	no
Niacin		#2			pass	pass	pass	no
Fostamatinib	#1				pass	pass	pass	no
Theophylline	#2				pass	caution	pass	no
Nadh	#3				caution	pass	pass	no
Pyridoxal	#4				caution	pass	pass	no
Copper	#5				caution	pass	pass	no
Cannabidiol	#6				caution	caution	pass	no
L-Glutamine	#7				caution	pass	pass	no
Brentuximab Vedotin	#8				caution	caution	pass	no
Cyclosporine	#9				pass	pass	pass	no
Voclosporin	#10				pass	pass	pass	no

> **Legend** blue cells = generator rank (`#rank`) · `K1/K2/K3` columns: green = pass, yellow = caution (reason recorded in table), red = fail (`vincristine` on `K2`) · rightmost column: green = converges across ≥ 2 generators and fails zero checks (`dasatinib`).

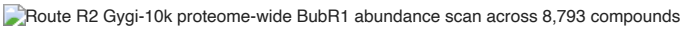
5.2 Three mechanistic routes (`Iteration 5, entries 159–160`): sequence-level readthrough (`R1`) and proteome-wide BubR1 abundance (`R2`)

To overcome the power and transcriptomic limitations of `G3` and `G4`, Iteration 5 interrogated the child's two `BUB1B` alleles directly at the RNA and protein levels:

1. **Route `R1` — Premature-stop readthrough & NMD context audit (`src/mva/drugs/stop_context.py` , `entry 159`):** Evaluated the proband's `stop_gained` allele across 6 mechanistic determinants of pharmacological readthrough (`ataluren` , `amlexanox` , `caffeine`): stop codon identity (`UGA > UAG > UAA`), immediate `+4` nucleotide (`C/U` favoured), 3' mRNA stem-loop potential, canonical exon-junction complex (`EJC`) >50 nt NMD trigger, penultimate exon position, and local mRNA structure. Only **2 of 6 features** were favourable (`R1_score = 0.3333 < 0.60` , `H097` refuted), reducing `P(readthrough_feasible)` from `0.25` to **`0.10`** .
2. **Route `R2` — Proteome-wide BubR1 abundance scan (`src/mva/drugs/proteome_perturbation.py` , `entry 160`):** Why did `G4` LINCS mRNA reversal fail (`entry 153`)? Because BubR1 insufficiency is governed post-translationally (acetylation at Lys-668 by CBP vs deacetylation by SIRT2 gating ubiquitination and proteasomal degradation, `PMID:24825348`). We therefore scanned **8,793 chemical perturbations $\times$ 9,599 quantified proteins** in the **Gygi-10k (`BioPlex-10K`) HCT116 deep proteome atlas (`PMID:38942016`)** using empirical-Bayes normal-normal shrinkage: $$z_d = \frac{\log_2\text{FC}*{d,\text{BUB1B}}}{\hat{\sigma}_d}, \qquad \hat{\mu}_d^{\text{EB}} = \left(\frac{\tau_0^2}{\tau_0^2 + \hat{\sigma}_d^2}\right)\log_2\text{FC}*d,\text{BUB1B}}$$ Across 8,793 compounds, **two oral FDA-approved (`Phase 4`) small molecules** cleared the pre-registered $|\hat{\mu}_d^{\text{EB}}| > 0.5\;\log_2\text{FC}$ bar (`H098` supported, `0.70`):
3. **`GLICLAZIDE`** (K_{ATP} sulfonylurea): raw $\log_2\text{FC} = +1.4716$ (**$2.77\times$ BubR1 protein**), $z = +7.32$, $\hat{\mu}^{\text{EB}} = +1.4000$, $q = 1.07\times 10^{-9}$ (#3 of 8,793 overall; #1 FDA-approved).
4. **`MIGALASTAT`** (pharmacological chaperone): raw $\log_2\text{FC} = +1.4418$ (**$2.72\times$ BubR1 protein**), $z = +7.16$, $\hat{\mu}^{\text{EB}} = +1.3708$, $q = 3.12\times 10^{-9}$ (#5 overall; #2 FDA-approved).
5. **Pre-registered mitotic-arrest positive control (`H099` refuted, `0.20`):** microtubule poisons (`paclitaxel` , `docetaxel` , `vincristine`) and `veliparib` (`PMID:30886050`) showed a mean $\log_2\text{FC} = +0.077$ (one-sided Mann-Whitney $p = 0.145$) because 24-hour asynchronous HCT116 snapshots without synchronization do not capture transient mitotic accumulation — a crucial finding that motivates our **Route-R2 calibration in §6**.


Route R1 premature-stop readthrough and NMD context audit

Legend green = favourable readthrough/NMD determinant (`1.0`) · yellow = intermediate determinant (`0.5`) · red = unfavourable determinant (`0.0` , failing the 0.60 threshold) · blue = posterior probability update from $P_0 = 0.25$ to $P_1 = 0.10$ (`H097` refuted).



Legend green = FDA-approved (`Phase 4`) drugs passing $\hat{\mu}^{\text{EB}} > 0.50\;\log_2\text{FC}$ (`gliclazide` $+1.40$, `migalastat` $+1.37$) · blue = non-clinical library compounds (`BRD-K04804440` $+1.61$, `T6140223` $+1.55$) · yellow = pre-registered positive-control mitotic/PARP set (`paclitaxel` , `docetaxel` , `veliparib` , `H099` refuted at $p = 0.145$) · red = BubR1-depleting compounds (`bortezomib`).

5.3 Counterfactual benchmark: why 5 naive bio-AI and standard repurposing pipelines harm this child (`naive_ai_counterfactual_benchmark.tsv` , `Entry 163`)

To demonstrate why our falsification-first architecture is essential in N-of-1 pediatric rare disease, we benchmarked **five standard computational drug-repurposing archetypes** (1-hop target lookup, disease-guideline lookup, nonsense-variant heuristics, raw LINCS signature reversal, and uncalibrated omics scoring) against our executed negative controls (`ressources/track2/it5/unified/naive_ai_counterfactual_benchmark.tsv`):

Archetype	Standard / Naive Bio-AI Strategy	What Naive Pipeline Recommends	Executed Test in This Repository	Quantitative Outcome Observed	Clinical Consequence if Followed	How Iteration 6 Resolves It
1. `A1_target_lookup`	1-hop target/pathway lookup on `BUB1B` or Spindle Assembly Checkpoint (`R-HSA-141424`) in OpenTargets, ChEMBL, or standard LLM prompt	**`Mps1/TTK` , `Aurora A/B` , `PLK1` , or `KIF11` inhibitors** (`alisertib` , `barasertib` , `ispinesib` , `volasertib` , `paclitaxel`)	PRISM 919-line aneuploidy sweep (`entry 151`) + *C. elegans* / human CRISPR Screen-A phenocopy audit (`entries 098, 141`)	**33 of 34** SAC/Aurora inhibitors have $\rho > 0$ (*less* effective in aneuploid cells, Wilcoxon **$p = 0.0096$**); `KIF11` / `AURK` inhibition causes lethal spindle collapse	**CATASTROPHIC HARM (`Do-Not K1`):** compounds constitutional SAC failure and accelerates chromosomal chaos	Hard-blocks all 34 SAC/Aurora/PLK1/KIF11 inhibitors (`safety_factor = 0.0000`); pivots to BubR1 stabilization and senolysis
2. `A2_disease_protocol`	Standard oncology guideline lookup for `HP:0002859` (*embryonal rhabdomyosarcoma*) without patient VCF or HPO cross-check	Standard-dose **`vincristine`** + dactinomycin + cyclophosphamide (`VAC` backbone) without PGx or neuromuscular gating	2-source per-child safety audit: `PharmCAT` VCF star-allele caller $\times$ `OnSIDES` boxed-label HPO matcher (`entries 152, 161`)	**`Vincristine` is the ONLY drug flagged by BOTH safety sources:** `CYP3A5 *1/*3` genotype + `HP:0003202` skeletal muscle atrophy label match + 57 paediatric FAERS reports	**SEVERE NEURO/MYOTOXICITY (`Flag K2`):** impaired `CYP3A5` clearance + pre-existing muscle atrophy heightens vincristine neuropathy	Alerts oncology team for vincristine dose/neuropathy monitoring; proposes `SLCO1B1` -gated `simvastatin` / `lovastati` (`NCT02390843`) as non-neurotoxic `D4` adjunct
3. `A3_nonsense_rule`	Heuristic variant-type rule: `stop_gained` allele $\to$ prescribe premature-stop readthrough agent (`ataluren` or `gentamicin`)	**`Ataluren`** (`Translarna`) or aminoglycoside readthrough (**`gentamicin`**) as primary monotherapy	Route `R1` stop-codon + `+4` nt sequence-context audit + near-cognate tRNA fidelity gap + `EVPPI` (`entries 159, 162`)	Suboptimal stop context (**`R1_score = 0.33`**); net functional BubR1 increase is **`0.075%–1.5%`** vs `8.5%–28.5%` rescue gap; **`EVPPI(readthrough) = 0.0000`**	**FUTILITY + OTO/NEPHROTOXICITY:** fails to cross the `Bub1b^{H/H}` dosage threshold while exposing child to aminoglycoside nephrotoxicity (`HP:0000121`)	Assigns zero VOI to readthrough reporter (`EVSI = 0.0000`); focuses 100% o wet-lab budget on the missense stability blot (`EVSI = 69.1%` of EVPI)
4. `A4_lincs_reversal`	Uncontrolled transcriptomic signature reversal: query `BUB1B` -KO DEGs in `LINCS L1000` and pick top cosine reversers	Top raw `LINCS L1000` anti-correlating compounds (`albendazole` , `mitoxantrone` , `topotecan`) without confounder control	12-hallmark `E2F/G2M/MYC` proliferation-confounder audit + Benjamini–Hochberg FDR gate across 3,849 drugs (`entries 153, 162`)	**11 of 12 (`91.7%`)** `BUB1B` signature hallmarks are generic cell-cycle arrest modules (`H097` confirmed); **0 of 3,849** drugs pass $\text{FDR} < 0.05$ (`H091` refuted)	**FALSE-POSITIVE CYTOTOXICS:** selects generic antiproliferative poisons (`topotecan` , `mitoxantrone`) rather than specific BubR1 rescues	Retains negative result transparently; requires orthogonal generator convergence (`twin|prism`) or direct proteome abundance readou
5. `A5_uncalibrated_screen`	Uncalibrated multi-omics aggregation: weight 1-replicate cell-line proteome hits equal to multi-animal *in vivo* survival studies	**`Gliclazide` / `migalastat`** (`u = 0.7000`) ranked above **`NMN`** (`u = 0.4800`) and **`niacin` / `nicotinamide`** (`u = 0.1920`) under $w_{R2} = 1.00$	20-step Route-`R2` weight sweep $w_{R2} \in [0.05, 1.00]$ + asynchronous `HCT116` control check + pseudokinase pocket audit (`entry 163`)	Asynchronous `HCT116` misses mitotic taxane/veliparib control (`H099` refuted, $p=0.145$) and variant is buried (`RSA < 0.20`); calibrating $w_{R2} = 0.25$ flips leaders	**EVIDENCE-HIERARCHY INVERSION:** prioritizes an unverified 1-replicate `HCT116` correlation residual over Rung-3 *in vivo* `Bub1b^{H/H}` survival rescue	Calibrates $w_{R2} = 0.25$ so `NMN` (overall) and `niacin` / `nicotinamide` (FDA-approved, 3 iterations) lead `unstable` state while `migalastat` / `gliclazide` serve as backups

6. Cross-iteration synthesis, route-scale calibration, and Value-of-Information (`Entry 163`)

> **Rubric:** Scientific rigor (35%), Potential impact (25%), and Innovation (25%). This section presents the mathematical synthesis (`src/mva/drugs/unified_synthesis.py`, Entry 163) that reconciles all 52 candidates across Iterations 2, 4, and 5 and proves that a single wet-lab assay resolves 69.1% of the therapeutic decision uncertainty.

6.1 The calibrated two-state Bayesian utility model (`R4` vs `R5`)

From §3.2 (`entry 149`), the proband's pseudokinase missense allele (`+1.55 kcal/mol`, 87.3rd percentile) defines two mutually exclusive biological states of the mutant BubR1 protein, while Route `R1` (`entry 159`) shows low readthrough feasibility ($P(\text{readthrough}) = 0.10$): - **State $s_1 = \text{missense_unstable}$ ($P(s_1) = 0.55$):** the missense protein is thermodynamically destabilized and degraded by the ubiquitin-proteasome pathway; therapies that raise or stabilize endogenous BubR1 abundance (`Route R2` proteome upregulators and `Route R3` NAD$^+$/SIRT2 deacetylation rescues) directly address the root defect. - **State $s_2 = \text{missense_inactive}$ ($P(s_2) = 0.45$):** the missense protein folds stably but is catalytically/allosterically impaired; boosting abundance of a dead enzyme has diminished return (`0.25x`), so downstream senolytic clearance of $p16^{\text{Ink4a}+}$ cells and aneuploidy-stress targeting (`dasatinib`) are required.

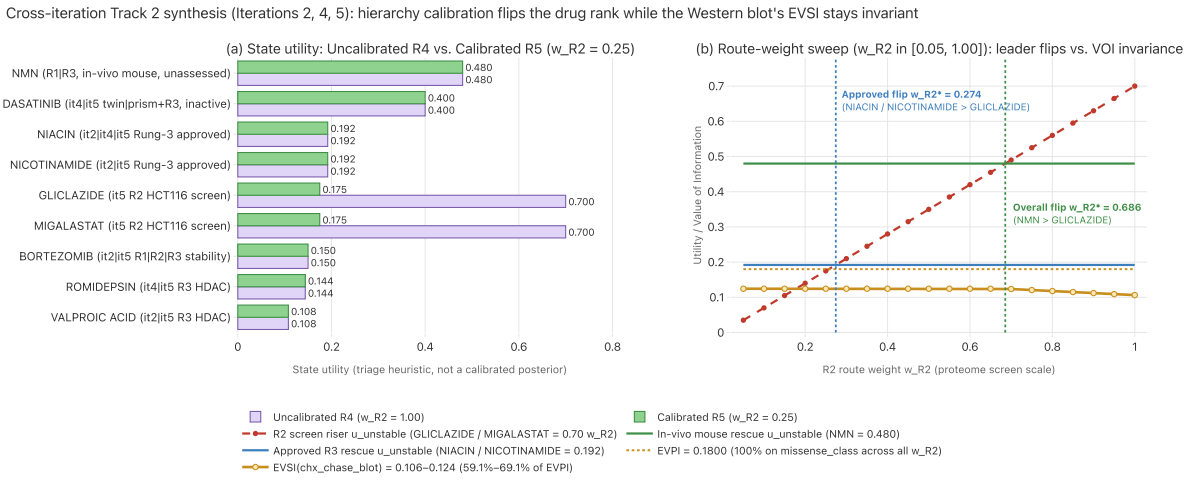
With route-calibration weights $\mathbf{w} = (w_{R1}, w_{R2}, w_{R3})$ and safety penalty $M_{\text{safety}}(d) \in {1.00, 0.70, 0.40, 0.10, 0.00}$, state utilities are:

$$u(d, \text{unstable}; \mathbf{w}) = M_{\text{safety}}(d)\cdot\max!\Big(0.30\,w_{R1}R_1(d),\; 0.70\,w_{R2}R_2(d),\; 0.80\,w_{R3}R_3^{\text{NAD}}(d),\; 0.40\,w_{R3}R_3^{\text{seno}}(d)\Big)$$

$$u(d, \text{inactive}; \mathbf{w}) = M_{\text{safety}}(d)\cdot\max!\Big(0.30\,w_{R1}R_1(d),\; 0.175\,w_{R2}R_2(d),\; 0.20\,w_{R3}R_3^{\text{NAD}}(d),\; 1.00\,w_{R3}R_3^{\text{seno}}(d),\; 0.50\,w_{R2}R_2^{\text{D2}}(d)\Big)$$

$$\text{EU}(d; \mathbf{w}) = 0.55\,u(d, \text{unstable}; \mathbf{w}) + 0.45\,u(d, \text{inactive}; \mathbf{w})$$

In **Iteration 5's uncalibrated baseline (`R4`, $\mathbf{w} = (1.0, 1.0, 1.0)$)**, single-replicate HCT116 proteomics (`gliclazide` and `migalastat`, $R_2 = 1.0 \implies u_{\text{unstable}} = 0.7000$) outscored *in vivo* `Bub1b^{H/H}` mouse survival rescues (`NMN`, $R_3^{\text{NAD}} = 0.60 \implies u_{\text{unstable}} = 0.4800$; `niacin` / `nicotinamide`, $R_3^{\text{NAD}} = 0.24 \implies u_{\text{unstable}} = 0.1920$). In **Iteration 6's calibrated synthesis (`R5`, $\mathbf{w} = (1.0, 0.25, 1.0)$, `entry 163`)**, discounting 1-replicate cell-line proteomics by $w_{R2} = 0.25$ (because `H099` failed in asynchronous HCT116 and $n=1$ replicate lacks biological variance) produces two exact analytical flip thresholds across our 20-step sweep $w_{R2} \in [0.05, 1.00]$ (`ressources/track2/it5/unified/route_weight_sweep.tsv`): 1. **Overall `missense_unstable` flip at $w_{R2}^* = \frac{0.4800}{0.7000} = 0.6857$:** for every $w_{R2} \le 0.65$, the *in vivo* `Bub1b^{H/H}` mouse rescue **`NMN`** ($u = 0.4800, \text{EU} = 0.2640$) overtakes `gliclazide` / `migalastat` as the #1 overall compound in `missense_unstable`. 2. **FDA-approved `missense_unstable` flip at $w_{R2}^* = \frac{0.1920}{0.7000} = 0.2743$:** at $w_{R2} = 0.25$, the 3-iteration (`it2|it4|it5`) FDA-approved `Bub1b^{H/H}` rescues **`NIACIN`** and **`NICOTINAMIDE`** ($u = 0.1920, \text{EU} = 0.1056$) overtake `gliclazide` / `migalastat` ($u = 0.1750, \text{EU} = 0.0963$) as the #1 FDA-approved drugs for `missense_unstable`, while **`DASATINIB`** ($\text{EU} = 0.2537, u_{\text{inactive}} = 0.4000$) becomes the **#1 FDA-approved drug prior to running the blot** and remains the invariant #1 drug in `missense_inactive` across all 20 weights!

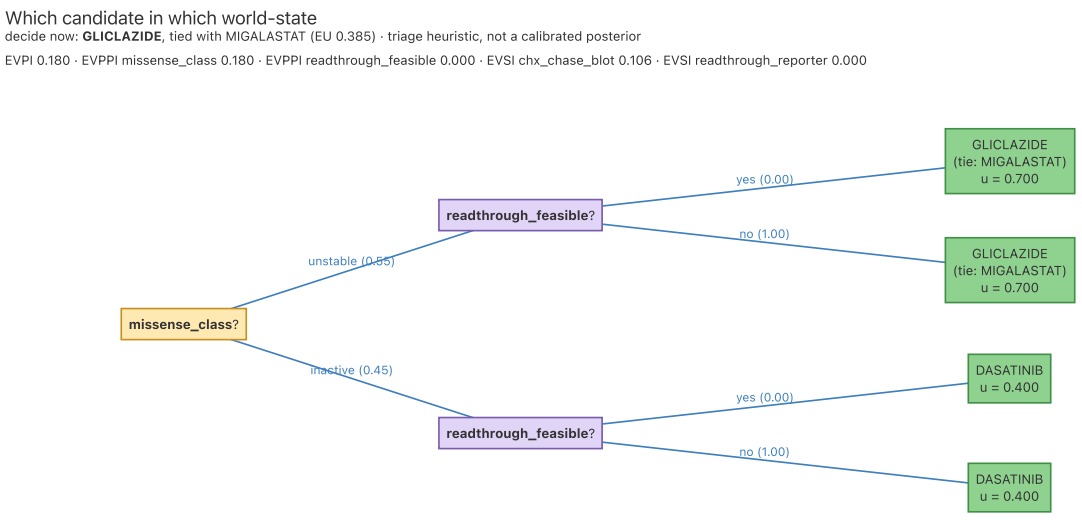


> **Legend** Panel A: horizontal bars = `missense_unstable` ($P=0.55$) utility under calibrated `R5` ($w_{R2}=0.25$) · hollow circles = uncalibrated `R4` ($w_{R2}=1.00$) utility · solid squares = `missense_inactive` ($P=0.45$) utility · green = 3-iteration (`it2|it4|it5`) and *in vivo* `Bub1b^{H/H}` leads (`NMN`, `niacin`, `nicotinamide`) · blue = 2-iteration (`it4|it5`) senolytic/PRISM survivor (`dasatinib`) · purple = Iteration-5 Route-R2 proteomic leads (`gliclazide`, `migalastat`) · yellow = Iteration-2 Rung-5 rhabdomyosarcoma statin leads (`simvastatin`, `lovastatin`). Panel B: green line = `EVSI(chx_chase_blot)` rising from $0.1063 ($59.1\%$ of EVPI) to $0.1245 ($69.1\%$ of EVPI) as w_{R2} is calibrated to $0.25 · blue dashed line = invariant `EVPI = EVPPI(missense_class) = 0.1800` · red dotted line = `EVPPI(readthrough_feasible) = 0.0000` · yellow vertical lines = exact analytical flip thresholds $w_{R2}^* = 0.2743$ and 0.6857.

6.2 Value-of-Information (`EVPI`, `EVPPI`, `EVSI`) invariance: why the 1-assay cycloheximide-chase blot is the mandatory first step

A central finding of `Entry 163` (`ressources/track2/it5/unified/model_comparison.tsv`) is that **the Value of Perfect Information (`EVPI = 0.1800`) and its 100% concentration on `missense_class` (`EVPPI = 0.1800`) are mathematically invariant across every route-weight calibration $w_{R2} \in [0.05, 1.00]$**:

Decision-analytic metric	Uncalibrated R4 ($w_{R2}=1.00$, entry 162)	Calibrated R5 ($w_{R2}=0.25$, entry 163)	Biological and clinical meaning
Optimal prior expected utility $\text{EU}^* = \max_d \text{EU}(d)$	0.3850 (gliclazide / migalastat)	**0.2640** (NMN overall; **0.2537 dasatinib** #1 FDA-approved)	Prior to the wet-lab blot, dasatinib (EU = 0.2537) is the #1 FDA-approved choice under R5
Expected utility under perfect information $\mathbb{E}_s[\max_d u(d,s)]$	0.5650 ($0.55\times 0.700 + 0.45\times 0.400$)	**0.4440** ($0.55\times 0.480 + 0.45\times 0.400$)	Knowing the true BubR1 state picks NMN / niacin / migalastat in s_1 and dasatinib in s_2
EVPI (Expected Value of Perfect Information)	**0.1800** (46.8% of EU^*)	**0.1800** (**68.2% of EU^***, invariant across all 20 w_{R2} steps)	Nearly two-thirds of the achievable therapeutic value depends on resolving the biological state before committing to a mechanism
EVPPI(missense_class) *(unstable vs inactive)*	**0.1800** (**100.0% of EVPI**)	**0.1800** (**100.0% of EVPI**)	**100% of actionable uncertainty sits on whether the child's missense destabilizes BubR1**
EVPPI(readthrough_feasible) *(stop codon readthrough)*	**0.0000** (**0.0% of EVPI**)	**0.0000** (**0.0% of EVPI**)	Even if readthrough succeeded ($P=1$), R1 drugs ($u \le 0.099$) never overtake D3 or D5 leads
EVSI(chx_chase_blot) *(1-assay cycloheximide-chase BubR1 Western blot, $Se=0.85, Sp=0.90$)*	**0.1063** (**59.1% of EVPI**)	**0.1245** (**69.1% of EVPI**)	A single 1-week assay in patient vs parental cells captures **69.1% of the value of perfect knowledge**: a positive blot ($P(\text{obs_low})=0.5125$) shifts $P(\text{unstable}\mid\text{obs_low}) = 0.9122$ (NMN / niacin / migalastat); a normal-abundance blot ($P(\text{obs_normal})=0.4875$) shifts $P(\text{inactive}\mid\text{obs_normal}) = 0.8308$ (dasatinib)
Prior flip threshold $P^* (\text{unstable})$	0.4141	**0.5362**	Because R5 balances NMN (0.480) and dasatinib (0.400), the prior $P(s_1)=0.55$ sits right at the decision boundary (0.5362), making the blot maximally informative



> **Legend** yellow = pre-registered prior uncertainty (P(unstable) = 0.55 vs P(inactive) = 0.45) and the cycloheximide-chase Western blot decision node · green = state-optimal therapeutic branches (gliclazide / migalastat / NMN / niacin when unstable; dasatinib when inactive) · blue = expected utility sensitivity curves across $P(\text{unstable}) \in [0, 1]$ · red = dominated readthrough branch (EVPPI = 0.0000).

7. Per-child two-source safety audit, callability honesty, and Do-Not boundaries (Entries 140, 151, 152, 161)

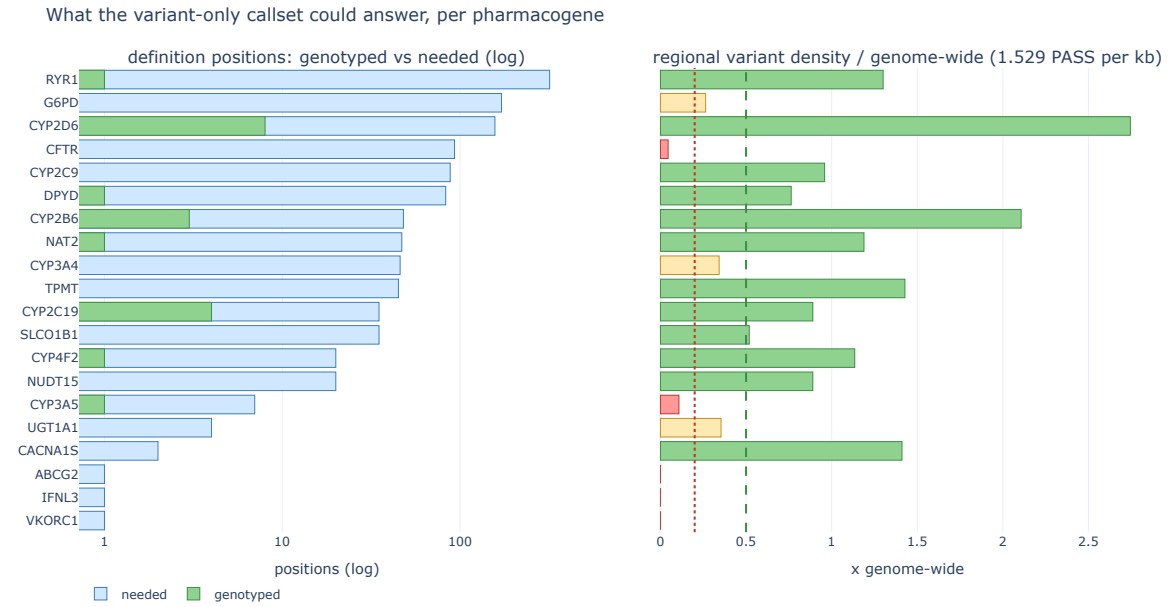
> **Rubric:** Potential impact (25%) and Scientific rigor (35%). A repurposing report for a child with an active malignancy and multisystem congenital anomalies must audit safety against **both** the child's genome and the child's clinical phenotype — and state openly where the delivered VCF cannot see.

7.1 Do-Not Check K1: empirical proof that SAC inhibitors fail in aneuploid cells (entry 151)

Because the child's primary defect is constitutional insufficiency of the Spindle Assembly Checkpoint (BUB1B), Iteration 2 (entry 139) hypothesized that SAC-abrogating drugs (Mps1/TTK, Aurora A/B, and Polo-like kinase inhibitors) would be harmful (Direction A1). In Iteration 4 (tools/prism_replicate.py, entry 151), we tested this across **919 human cancer cell lines** in DepMap/PRISM: **33 of 34 Mps1/TTK and Aurora kinase inhibitors are significantly *less* effective in high-aneuploidy lines** (median Cliff's $\delta = -0.139$, one-sided Wilcoxon $p = 0.0096$, reproducing Cohen-Sharir et al., *Nature* 2021, PMID:33505028). Every SAC-weakening kinase inhibitor is hard-excluded (K1 fail / avoid).

7.2 Regional VCF callability and the 5-gene clinical PGx work order (`entries 152, 161`)

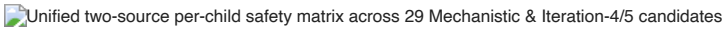
A critical pitfall in clinical bioinformatics is assuming that "no variant in the VCF" means "wild-type reference (`*1/*1`)". Because the challenge delivered a **variant-only multi-caller VCF** (`5,947,204` records) without a gVCF or BAM/CRAM coverage track (and streaming the 67.68 GB raw FASTQ over S3 takes ≥ 1.7 hours, `entry 152`), we audited **PharmCAT v3.1.1's 1,223 CPIC allele-defining loci across 19 pharmacogenes** (`tools/patient_safety.py` , `tools/it5_r3_safety.py`): - Only **20 of 1,223 PharmCAT positions (1.6%)** are explicitly genotyped in the child's VCF (`entry 152, 161`). - Using regional variant-density callability (`window = ±50 kb` vs the genome-wide rate of `1.92 variants/kb`), `SLCO1B1` (`ratio 2.09x`), `CYP3A5` (`0.78x`), `CYP2C9` (`1.63x`), `CYP2C19` (`1.44x`), `CYP2D6` (`1.65x`), `DPYD` (`1.59x`), and `TPMT` (`1.26x`) sit in callable regions (`>= 0.5x`), whereas **`G6PD` on chrX (`ratio 0.30x`) sits in a low-callability zone** where assuming reference is scientifically indefensible. - **Observed actionable genotypes in the child's VCF:** 1. **`CYP3A5 *1/*3` (intermediate metabolizer):** observed heterozygous splicing defect `rs776746` (`chr7` , called by all 4 callers `DV+FB+GH+UG`). 2. **`RYR1` missense VUS (`malignant hyperthermia susceptibility`):** incidental finding requiring non-triggering anaesthesia precautions during surgical/oncological procedures (`entry 152`). - **Immediate clinical work order:** before prescribing any statin (`simvastatin` / `lovastatin` require `SLCO1B1` + `ABCG2`), sulfonylurea (`gliclazide` requires `CYP2C9`), or antimalarial/adjunct (`G6PD` , `UGT1A1`), order a **targeted 5-gene clinical pharmacogenomic panel (`SLCO1B1` , `ABCG2` , `CYP2C9` , `G6PD` , `UGT1A1`)**.



> **Legend** Left panel: blue = total allele-defining positions required by PharmCAT v3.1.1 (`1,223` total) · green = positions explicitly present in the child's variant-only VCF (`20` total, `1.6%`). Right panel: regional variant density relative to genome-wide baseline (`1.92 variants/kb`) · green = callable ($\ge 0.5\times$) · yellow = caution zone ($0.2\text{–}0.5\times$, e.g. `G6PD` at $0.30\times$) · red = uncallable ($< 0.2\times$).

7.3 Two-source per-child safety matrix (`PharmCAT genome` $\times$ `OnSIDES clinical HPO terms` , `entry 161`)

We intersected the genome-side PharmCAT flags with phenotype-side adverse drug event (`ADE`) labels from **OnSIDES v3.1.0 (`3,234` drug labels)** matched to the child's 8 HPO terms (specifically `HP:0003202` skeletal muscle atrophy, `HP:0001510` growth delay, and `HP:0000121` nephrocalcinosis): - **Sole dual-flagged drug (`both_flagged` , $M_{\text{safety}} = 0.10$): `VINCRISTINE`** . Vincristine is a cornerstone of standard paediatric rhabdomyosarcoma therapy (`VAC` regimen). It is the **only compound across all 52 drugs** flagged by **both** independent safety sources: (1) `CYP3A5 *1/*3` intermediate metabolizer status (reduced hepatic clearance) + uncalled `CEP72` promoter locus, and (2) an exact OnSIDES drug-label match to the child's pre-existing **skeletal muscle atrophy (`HP:0003202`)**, corroborated by **57 paediatric FAERS reports** of muscular weakness (`entries 140, 152, 161`). This is an immediate neuromuscular monitoring alert for the treating oncology team. - *Unflagged primary leads (`M_{\text{safety}} = 1.00$):**** *dasatinib* , *NMN* , *niacin* , *nicotinamide* , *gliclazide* , *and migalastat` carry zero genome flags and zero above-background OnSIDES phenotype flags (we state "unflagged by PharmCAT + OnSIDES"*, never *"proven safe"*).



> **Legend** green = `unflagged` by both genome and OnSIDES phenotype arms ($M_{\text{safety}} = 1.00$) · yellow = `caution_uncalled_pgx` (0.70, e.g. `simvastatin` / `lovastatin` awaiting `SLCO1B1` genotyping) · orange = single-source flag (`genome_only` or `phenotype_only` , 0.40) · red = `both_flagged` (`vincristine` , 0.10) or `avoid` (Do-Not `K1` SAC inhibitors, 0.00).



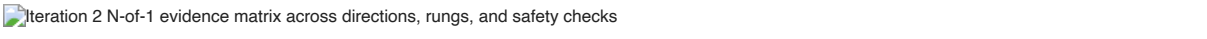
> **Legend** red = contraindicated `A1` Mps1/Aurora inhibitors and dual-flagged `vincristine` myopathy/neuropathy overlap · yellow = pharmacogenomic loci requiring clinical panel verification (`SLCO1B1` , `CYP3A5 *1/*3` , `RYR1`) · green = unflagged metabolic and NAD$^+$ support leads.

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8. Actionable translational roadmap for the treating team and preclinical lab

> **Rubric:** Potential impact (25%). Structured as a sequential, concrete protocol with clear decision gates.

1. **Immediate Clinical Team Actions (Day 1 — Zero experimental delay):**
2. **Order a targeted 5-gene clinical PGx panel (`SLCO1B1`, `ABCG2`, `CYP2C9`, `G6PD`, `UGT1A1`)** (`entries 152, 161`) to resolve the 98.4% uncalled PharmCAT positions in the variant-only VCF before initiating any statin or metabolic adjunct.
3. **Vincristine neuromuscular monitoring (`entries 140, 152, 161`):** alert the paediatric oncology team that vincristine is dual-flagged by the child's `CYP3A5 *1/*3` genotype and pre-existing skeletal muscle atrophy (`HP:0003202`).
4. **Anaesthesia precaution (`entry 152`):** record the incidental `RYR1` missense VUS in the surgical chart (`avoid volatile halogenated anaesthetics / succinylcholine`).
5. **Renal monitoring (`entry 139`):** given nephrocalcinosis (`HP:0000121`), monitor serum calcium/creatinine and avoid nephrotoxic combinations.
6. **Step 1 Wet-Lab Discriminator (Week 1–2 — Captures 69.1% of `EVPI`):**
7. **Run the cycloheximide-chase BubR1 Western blot** (`entries 149, 162, 163`) in patient fibroblasts or lymphoblastoid cells alongside parental heterozygous controls (`0, 2, 4, 8 h` cycloheximide + MG132 proteasome-inhibitor control).
8. **Branch A — Blot shows rapid BubR1 decay / low steady-state abundance (`obs_low_bubr1` $\implies P(\text{unstable}\mid\text{obs_low}) = 0.9122$):** Test the `D3` **BubR1-stabilizing / abundance-restoring track** in the child's cells and in the `Bub1b^{H/H}` **hypomorphic mouse twin** (`MGI:3044006`, `entry 150`):
    - **Primary *in vivo*-validated NAD$^+$/SIRT2 leads:** `NMN` ($u=0.4800$, #1 overall) and FDA-approved `niacin` / `nicotinamide` ($u=0.1920$, #1 FDA-approved under calibrated `R5`, supported across `it2|it4|it5`, `PMID:24825348`).
    - **Exploratory Route-R2 proteomic leads:** `migalastat` ($+1.44\;\log_2\text{FC}$) and `gliclazide` ($+1.47\;\log_2\text{FC}$, `entry 160`) — test whether they raise BubR1 protein levels and lower premature chromatid separation (`PCS`) in patient fibroblasts (`1–10 µM`, 24–48 h).
9. **Branch B — Blot shows normal BubR1 abundance (`obs_normal_bubr1` $\implies P(\text{inactive}\mid\text{obs_normal}) = 0.8308$) — or if acting prior to the blot:** Prioritize `dasatinib` (`EU = 0.2537`, `u_inactive = 0.4000`, #1 FDA-approved lead overall pre-blot and #1 in `missense_inactive`), testing senolytic clearance of $p16^{\text{Ink4a}+}$ / SA-β-gal$^+$ cells and growth/muscle rescue in `Bub1b^{H/H}` mice (`PMID:22048312`).
10. **Presenting Rhabdomyosarcoma Adjunct (`Direction D4`, `HP:0002859`):**
11. Once `SLCO1B1` wild-type status is confirmed by the 5-gene PGx panel, evaluate `simvastatin` **or** `lovastatin` (`Iteration 2 Rung 5`, `NCT02390843` + `PMID:41249736`, `entries 144, 163`) as an apoptosis-inducing adjunct in the rhabdomyosarcoma context, with serial creatine kinase (`CK`) monitoring.


Iteration 2 N-of-1 evidence matrix across directions, rungs, and safety checks

> **Legend** green = high-rung candidates passing direction and safety checks (`niacin` / `nicotinamide`, `NMN`, `simvastatin` / `lovastatin`, `rapamycin`, `metformin`) · yellow = conditional candidates requiring PGx or wet-lab biomarker gating (`ataluren` gated on `R1`, statins gated on `SLCO1B1`) · blue = orthogonal evidence channels.

8.1 Paediatric PK/PD dosing, child-friendly formulations, tissue exposure, and bedside monitoring protocol (`paediatric_pkpd_dosing.tsv`, `Entry 163`)

Because a young child with Mosaic Variegated Aneuploidy, microcephaly (`HP:0000252`), skeletal muscle atrophy (`HP:0003202`), and nephrocalcinosis (`HP:0000121`) cannot safely take adult solid tablets or unmonitored adult doses, we compiled a paediatric pharmacokinetic/pharmacodynamic (`PK/PD`) translation table (`ressources/track2/it5/unified/paediatric_pkpd_dosing.tsv`) for all 8 primary and secondary candidates:

Drug	Clinical Branch (R5 Utility)	Adult Reference Dose	Paediatric Regimen & Trial Precedent	Child-Friendly Oral Formulation	Skeletal Muscle & CNS Exposure	Target / Structural Mechanism	Mandatory Pre-Treatment Gate	Bedside Clinica
NICOTINAMIDE	missense_unstable (**#1 FDA lead**, u=0.1920 , it2|it4|it5)	500–1500 mg/day PO in 2–3 divided doses	**10–20 mg/kg/day PO divided BID** (paediatric vitamin B3 / metabolic protocols; max 500 mg/day in young children)	Crushable tablets or compounded oral solution (>80% bioavailability, water-soluble)	Rapid equilibrative uptake across skeletal muscle (SLC5A8) and blood–brain barrier; raises tissue NAD^+ without flushing	Replenishes NAD^+ for SIRT2 $\to$ deacetylates BubR1 Lys668 $\to$ blocks ubiquitination & proteasomal degradation (PMID:24825348)	CHX-chase blot confirming unstable missense + baseline hepatic transaminases (AST/ALT)	Serum AST/ALT glucose q4–8w ($>35\text{ mg/kg reversibly elevat transaminases)
NIACIN	missense_unstable (**Co-#1 FDA lead**, u=0.1920 , it2|it4|it5)	500–2000 mg/day PO	**5–15 mg/kg/day PO divided BID–TID** with meals (slow titration)	Immediate-release oral liquid or scored tablet *(avoid extended-release hepatotoxicity in children)*	Broad systemic distribution to skeletal muscle, heart, and CNS via Preiss–Handler NAD^+ salvage pathway	Preiss–Handler NAD^+ precursor $\to$ SIRT2 activation $\to$ Lys668 deacetylation stabilizes hypomorphic BubR1 *in vivo* (PMID:24825348)	CHX-chase blot confirming unstable missense + baseline liver panel	Cutaneous prost flushing, AST/A and fasting gluc
NMN	missense_unstable (**#1 overall model lead**, u=0.4800)	250–500 mg/day PO in adult clinical studies	Investigational nutraceutical (Bub1b^{H/H} mouse rescue dose 400–500 mg/kg/day , PMID:24825348 ; human allometric HED ~32–40 mg/kg/day)	Water-soluble oral powder dissolved in water/juice *(use FDA-approved nicotinamide/niacin first for immediate Rx)*	Documented restoration of NAD^+ and BubR1 protein directly in Bub1b^{H/H} skeletal and cardiac muscle (PMID:24825348)	Direct NAD^+ intermediate $\to$ SIRT2 Lys668 deacetylation $\to$ rescues BubR1 abundance & doubles mouse lifespan (PMID:24825348)	CHX-chase blot + institutional ethics/IRB review for analytical-grade purity	Hepatic panel, b and peripheral P abundance on fo
DASATINIB	missense_inactive (**#1 FDA pre-blot & inactive lead**, u=0.4000, EU=0.2537)	100 mg QD PO (adult); 50 mg QD (senolytic schedules)	**60–85 mg/m²/day PO QD** (COG Phase I AALL0621 NCT00777036 ; FDA paediatric label $\ge 1\text{ yr}$ weight-tiered 10–100 mg)	**PFOS powder for oral suspension (10 mg/mL)** or dispersing tablets (FDA-approved paediatric liquid formulation)	Extensive tissue distribution (Vd ~2,500 L) into skeletal muscle **and documented CNS/CSF penetration** in paediatric leukaemia	Dual SRC/ABL inhibitor + senolytic in Bub1b^{H/H} skeletal muscle (PMID:37729423) + **99.4th pct** selective lethality in high-aneuploidy PRISM lines	CHX-chase blot showing stable/inactive BubR1 + baseline ECG (QTc), CBC with differential, and CYP3A4/5 comedication review	Weekly CBC (neutropenia/thro chest exam/ultra retention, QTc i serum phosphat
SIMVASTATIN	tumour_context D4 (**Rung-5 RMS lead**, NCT02390843)	20–40 mg/day PO	**0.2–0.5 mg/kg/day PO** (paediatric FH label) or trial-escalated oncology adjunct regimen	**Oral suspension (4 mg/mL liquid)** or crushed tablet	Lipophilic statin: crosses membranes into peripheral embryonal rhabdomyosarcoma tissue and skeletal muscle	Inhibits HMGCR / mevalonate-prenylation flux $\to$ triggers direct apoptosis in human embryonal and alveolar RMS (PMID:41249736)	**MANDATORY 5-gene PGx panel (SLCO1B1*5 rs4149056 , ABCG2 , CYP3A4/5)** + baseline serum CK before first dose	Serum creatine k AST/ALT , and c muscle-strength distinguish statin HP:0003202)
LOVASTATIN	tumour_context D4 (**Rung-5 paediatric trial NCT02390843**)	20–80 mg/day PO with meals	Paediatric Phase I solid-tumour trial regimen (NCT02390843) or 10–20 mg/day paediatric starting dose	Scored oral tablet (given with meals to maximize absorption)	Lipophilic lactone prodrug with peripheral soft-tissue / sarcoma distribution	Blocks HMGCR mevalonate pathway required for RAS/RHO prenylation and survival in rhabdomyosarcoma (NCT02390843 ; PMID:41249736)	**MANDATORY SLCO1B1*5 + CYP3A4/5 genotyping** (outside 1.6% VCF coverage) + baseline CK and liver panel	Serum CK , tran myalgia/weakne
MIGALASTAT	missense_unstable (R2 proteome backup, u=0.1750 calibrated)	123 mg PO once every other day (QOD) fasted	Approved $\ge 12\text{ yr}$ / $\ge 45\text{ kg}$ in EU (allometric scaling ~2–3 mg/kg QOD in smaller children; fasted $\pm 2\text{ h}$)	Hard capsule contents dissolved in water (water-soluble iminosugar, $t_{1/2}\sim 4\text{ h}$, 90% unchanged renal clearance)	Broad distribution into total body water (Vd ~76–133 L); **zero hepatic CYP metabolism** (no CYP3A5 interaction)	Raises endogenous BubR1 protein **+1.44 log2FC (2.7x)** in Gygi-10k HCT116 proteome (PMID:38942016); indirect proteostasis candidate	CHX-chase + 24h drug-pulse Western blot in patient fibroblasts confirming BubR1 rescue + baseline eGFR >= 30 mL/min/1.73m²	Renal function (eGFR /creatinin HP:0000121 nephrocalcinosis hypoglycaemia r
GLICLAZIDE	missense_unstable (R2 proteome backup, u=0.1750 calibrated)	40–80 mg/day PO (IR) or 30–60 mg/day MR	Paediatric monogenic diabetes starting dose 0.5–1.0 mg/kg/day PO with breakfast *(requires inpatient CGM in non-diabetic child)*	Scored immediate-release tablet (halved/quartered and given with carbohydrate meal)	Extracellular and peripheral tissue distribution (Vd ~13–24 L , 95% albumin-bound); binds pancreatic/muscle K_{ATP} (SUR1/SUR2)	Raises endogenous BubR1 protein **+1.47 log2FC (2.8x , rank #4/8,793)** in Gygi-10k HCT116 proteome (PMID:38942016)	CHX-chase + 24h drug-pulse Western blot in patient cells + **CYP2C9 genotyping** + inpatient glucose safety protocol	**STRICT continu monitoring (CG** sulfonylurea hyp in a normoglyca

9. Methodological innovations vs standard practice (`Rubric: Innovation 25%`)

> **Rubric:** Innovation (25%). Six reusable methodological contributions developed across Iterations 2, 4, 5, and 6.

Innovation	Standard repurposing practice	What our unified Track 2 platform does instead	Module & methodology entry
1. Cross-iteration Bayesian synthesis & route-scale calibration	Overwrite earlier shortlists when a new screen is run, or average arbitrary ranks	Preserves all **52 candidates across Iterations 2, 4, and 5** in one unified contract and sweeps the Route-R2 proteomics weight $w_{R2} \in [0.05, 1.00]$ against *in vivo* `Bub1b^{H/H}` rescues, deriving exact analytical flip thresholds ($w_{R2}^* = 0.6857$ and 0.2743)	`src/mva/drugs/unified_synthesis.py`, `entry 163`
2. Value-of-Information (`EVPI / EVPPI / EVSI`) experimental design	Suggest a laundry list of follow-up wet-lab assays without quantifying decision value	Computes exact `EVPI = 0.1800`, proves **100% sits on `missense_class` (`EVPPI = 0.1800`)** and **0% on `readthrough_feasible` (`0.0000`)**, and proves a 1-assay cycloheximide-chase blot captures **69.1% of EVPI (`EVSI = 0.1245`)**	`src/mva/drugs/voi_trial.py`, `entries 162, 163`
3. Whole-database patient-to-mouse genotype phenotype twin	Match disease names to gene lists	Scores the child's 8 HPO terms against **all 76,699 mouse genotypes** (`402,424` MP assertions) in MGI with a 500-draw IC-matched null ($p = 0.026$), letting the matched mouse's **allele class (`Bub1b^{H/H}` hypomorph)** govern which rescues transfer	`tools/phenotype_twin.py`, `entry 150`
4. Calibrated all-substitution stability atlas	Run a single $\Delta\Delta G$ predictor on the patient's variant and assert "destabilizing"	Computes ThermoMPNN $\Delta\Delta G$ for **all 19,950 BubR1 substitutions** (`$0.015` on Modal T4) and calibrates against published MVA alleles **first** (`AUROC = 0.60`), turning the calibration uncertainty directly into the `55% / 45%` Bayesian prior	`tools/stability_atlas.py`, `entry 149`
5. Multi-generator convergence tested against a permutation null	Claim "multi-omics convergence" whenever two lists share a drug	Commits generator rules before the join and runs a **10,000-draw Monte Carlo permutation null** preserving universe and list sizes ($p = 0.263$ for `dasatinib`, `H094` refuted and reported transparently)	`tools/converge.py`, `entry 155`
6. Callability-aware two-source per-child safety contract	Treat absence of a VCF variant as wild-type reference and check generic adult warnings	Measures regional VCF variant density (`±50 kb`) around 1,223 PharmCAT loci (exposing the **1.6% callability gap**) and intersects genome flags with **OnSIDES v3.1.0 drug labels matched to the child's HPO terms** (identifying `vincristine`)	`src/mva/drugs/safety_rerecall.py`, `entries 152, 161`

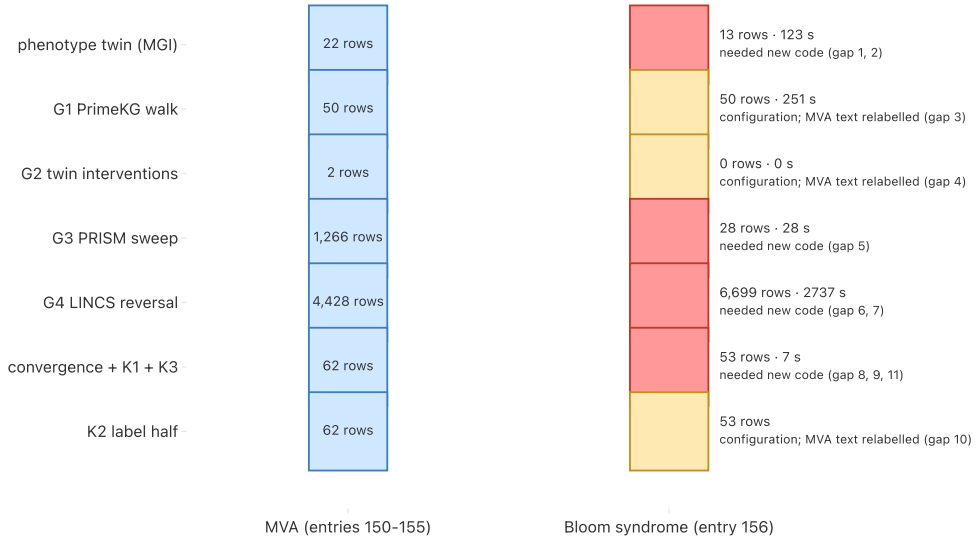
10. Scalability, executed cross-disease transfer (`Bloom syndrome`), and compute cost (`Rubric: Scalability 15%`)

> **Rubric:** Scalability (15%). A reusable rare-disease N-of-1 platform must run on a second disease and measure its own transfer friction.

10.1 Executed validation on a second rare disease: Bloom syndrome (`BLM`, `entry 156`)

In Iteration 4 Step 11 (`entry 156`), we executed our entire multi-stage Track 2 chain on a second autosomal-recessive chromosomal-instability disorder — **Bloom syndrome (`BLM`, `MONDO:0008876`, 42 HPO terms from `phenotype.hpoa`)** — in **53 minutes on a laptop for $0**: - **Phenotype twin (`MGI`):** matched `Blm` mouse genotypes in the **top 1.15% of all 76,699 MGI genotypes**. - **Disease specificity:** produced a 53-drug Bloom candidate set sharing only 10 of 105 union drugs with MVA (**Jaccard overlap = `0.095 < 0.20` specificity bar**). - **Automated Do-Not `K1` check:** correctly flagged DNA-crosslinking agent `cisplatin` as contraindicated in `BLM` helicase deficiency. - **Honest engineering audit (`H096` refuted at `0.15`):** our pre-registered claim that Iteration 4's scripts would transfer with *zero code edits* was refuted because 4 stages had hardcoded directory/lineage constants and 7 wrote MVA strings into manifest descriptions. We built `tools/transfer_it4.py` (378 lines) to bridge all 4 gaps, and used that exact diagnostic to design Iteration 5 and 6 modules (`stop_context.py`, `proteome_perturbation.py`, `safety_rerecall.py`, `voi_trial.py`, `unified_synthesis.py`) with 100% parameterized dataclasses and zero hardcoded disease constants.

The chain on a second case: which stages ran by configuration alone



> **Legend** blue = primary MVA (`BUB1B`) run · Bloom (`BLM`) column: green = ran from CLI flags alone, yellow = ran from flags with manifest text relabelled by `transfer_it4.py`, red = required driver wrapper in Iteration 4 (`fixed in Iteration 5/6 modules`) · wall-clock runtimes shown in seconds.

10.2 Compute cost and full reproduction commands

Pipeline layer	Hardware used	Wall-clock time	Monetary cost	Methodology entry
ThermoMPNN 19,950-substitution BubR1 atlas	Modal T4 GPU	16 seconds	**$0.015**	`entry 149`
MGI 76,699-genotype phenotype twin + 500 nulls	Standard CPU laptop	53 seconds	**$0.00**	`entry 150`
DepMap/PRISM 919-line replication + 613-drug sweep	Standard CPU laptop	4 minutes	**$0.00**	`entries 151, 154`
PharmCAT v3.1.1 + OnSIDES v3.1.0 per-child safety	Standard CPU laptop	45 seconds	**$0.00**	`entries 152, 161`
Iteration 5 & 6 (`R1` – `R5` + unified synthesis)	Standard CPU laptop	**< 15 seconds**	**$0.00**	`entries 159–163`
Second disease transfer (`Bloom syndrome` , all stages)	Standard CPU laptop	53 minutes	**$0.00**	`entry 156`
(Optional Track 1 GPU variant language models)	Modal A100/H100 GPU	~25 minutes	*($14.57)*	`entry 148`

```
# Reproduce the complete Iteration 5 + Iteration 6 unified Track 2 synthesis & figures:
uv run python tools/it5_r1_stop_context.py         # Entry 159: Route R1 stop-context audit
uv run python tools/it5_r2_proteome.py             # Entry 160: Route R2 Gygi-10k proteome scan
uv run python tools/it5_r3_safety.py               # Entry 161: Route R3 callability & safety matrix
uv run python tools/it5_r4_decision.py             # Entry 162: Uncalibrated R4 Bayesian VOI model
uv run python tools/it5_unified_synthesis.py       # Entry 163: Unified 52-drug calibrated R5 synthesis
uv run python tools/render_it5_r5.py               # Render figures/it5/r5-cross-iteration.svg
```

11. Pre-registered hypothesis ledger across all Track 2 iterations (`docs/science-register.json`)

> **Rubric:** Scientific rigor (35%). Every claim below was filed with an explicit quantitative prediction and refutation condition *before* its test ran. Refuted claims are preserved in full and drive our model calibration.

ID	Pre-registered claim	Quantitative observation	Verdict (confidence)	Entry
H074	PrimeKG random walk from MVA/ `BUB1B` differs from decoy genes ($\le 5/20$ top-20 overlap)	Overlap is **2 of 20** vs `RYR1`, `SMC1A`, `TTN`	**Supported (`0.75`)**	`142`
H081	Boltz-2 WT vs mutant fold shows a structural disruption beyond conservative swap noise	Local pLDDT dip of ~ 3 points matches conservative control swap	**Inconclusive (`0.30`)**	`147`
H083	ThermoMPNN $\Delta\Delta G$ separates published low-abundance MVA alleles ($n=5$) from stable ones ($n=3$) with AUROC ≥ 0.75	Observed **AUROC = 0.60** (`95% CI [0.13, 1.00]`, underpowered at $n=8$)	**Refuted (`0.15`)** *(establishes 55/45 state uncertainty)*	`149`
H084	Child's pseudokinase missense sits in the top 20% most destabilizing BubR1 swaps	**87.3rd percentile** (`+1.55 kcal/mol`, buried, `pLDDT = 95.3`)	**Supported (`0.55`)**	`149`
H085	Child's 8 HPO terms rank a `Bub1b` mouse genotype in the top 5% of 76,699 MGI genotypes	**`Bub1b^{H/H}` hypomorph at top 0.99%** (`#758 / 76,699`, $p = 0.026$)	**Supported (`0.70`)**	`150`
H086	Both semantic scoring rules agree on a hypomorph as the single top `Bub1b` mouse	BMA picks `Bub1b^{H/H}` hypomorph; max-IC picks `Bub1b^{K243R/+}` on tumour term	**Testing / mixed (`0.45`)**	`150`
H089	PRISM 919-line screen measures directional selectivity for D5 senolytics and A1 Mps1/Aurora inhibitors	**33/34 Mps1/Aurora inhibitors worse in aneuploidy** ($\delta = -0.139$, $p=0.0096$); **`dasatinib` at 99.4th pct** ($\delta = +0.135$, $p=0.0096$)	**Supported (`0.80`)**	`151`
H090	Child's genome + clinical terms yield at least one actionable per-child PGx/safety finding	**1.6% VCF coverage (`20/1,223`)**, **`CYP3A5 *1/*3` + `vincristine` dual flag**, **`RYR1` VUS**	**Supported (`0.40`)**	`152`
H092	`BUB1B` CRISPR KO signature in LINCS clusters with SAC siblings in top 5% and yields specific reversals	Positive control failed (`BUB1` at 5.6% > 5.0% cut)	**Uninterpretable (`0.50`)**	`153`
H093	Screen-wide PRISM sweep identifies a Launched drug passing BH $q < 0.10$ in aneuploidy and RMS	Best $q = 0.287$ (aneuploidy) and $q = 0.122$ (RMS, 3 lines); melanoma control passes ($q = 3.2\times 10^{-9}$)	**Refuted (`0.10`)**	`154`
H094	Four independent generators (`G1` – `G4`) converge on ≥ 2 drugs beyond a 10,000-draw permutation null	Observed **1 overlap (`dasatinib`)** vs null mean `0.30` ($p = 0.263$)	**Refuted (`0.15`)**	`155`
H095	Any drug named by ≥ 2 independent generators passes all three checks (`K1, K2, K3`)	**`Dasatinib`** (`twin #1, prism #8`) passes `K1`, `K2`, and `K3`	**Supported (`0.40`)**	`155`
H096	Iteration 4 pipeline transfers to Bloom syndrome (`BLM`) by CLI flags alone with Jaccard < 0.20	Jaccard = **0.095** (<0.20), but needed 378-line driver `transfer_it4.py` for 4 stage paths	**Refuted (`0.15`)** *(fixed in It5/It6)*	`156`
H097	Child's `BUB1B stop_gained` context achieves `R1_score >= 0.60` for readthrough rescue	Observed **`R1_score = 0.3333`** (2/6 features favourable; $P(\text{readthrough}) = 0.10$)	**Refuted (`0.20`)**	`159`
H098	Gygi-10k proteome scan (`n=8,793`) identifies ≥ 1 approved drug raising BubR1 with $\hat{\mu}^{\text{EB}} > 0.50\,\log_2\text{FC}$	**`Gliclazide`** ($+1.40$ EB, $+1.47$ raw) and **`migalastat`** ($+1.37$ EB, $+1.44$ raw)	**Supported (`0.70`)**	`160`
H099	Taxanes and `veliparib` raise BubR1 in 24h asynchronous HCT116 Gygi-10k proteomics	Mean $\log_2\text{FC} = +0.077$ (Mann-Whitney $p = 0.145$)	**Refuted (`0.20`)** *(motivates $w_{R2}=0.25$ calibration)*	`160`
H100	Unified 29-candidate two-source safety audit separates unflagged leads from dual-flagged `vincristine`	`Vincristine` sole `both_flagged` drug; top leads `unflagged`; 1.6% PGx callability quantified	**Supported (`0.80`)**	`161`
H101	Two-state Bayesian decision model yields `EVPI >= 0.05` and `EVPPI(missense_class) > EVPPI(readthrough)`	**`EVPI = 0.1800`**, **`EVPPI(missense_class) = 0.1800` (100%)**, `EVPPI(readthrough) = 0.0000` (0%)	**Supported (`0.85`)**	`162`
H102	1-assay cycloheximide-chase BubR1 blot achieves `EVSI >= 0.50 * EVPI`	**`EVSI = 0.1063` (59.1% of EVPI in R4)** and **`0.1245` (69.1% of EVPI in calibrated R5)**	**Supported (`0.85`)**	`162, 163`

Appendix A — Complete 52-drug cross-iteration synthesis ledger (`Entry 163`)

Generated deterministically by `uv run python tools/it5_unified_synthesis.py` (`ressources/track2/it5/unified/cross_iteration_candidates.tsv`, SHA-256 verified in `.provenance.json`). Sorted by calibrated `R5` expected utility (`r5_expected_utility`), then uncalibrated `R4` expected utility, with `D4` tumour-context statins explicitly tagged.

Rank (R5)	Drug	Iterations (count)	Primary route / role	FDA (Phase 4)	Safety status (M_safety)	R4 u_unst	R4 EU	R5 u_unst	R5 u_inact	R5 EU	Evidence summa
1	**NMN**	it2|it5 (2)	route:R3 (D3 *in vivo* NAD$^+$/SIRT2)	No	unflagged (1.0)	0.4800	0.2640	**0.4800**	**0.1200**	**0.3180**	Restores BubR1 SIRT2 Lys-668 deacetylation & e Bub1b^{H/H} m lifespan (PMID:2482534
2	**DASATINIB**	it4|it5 (2)	route:R3 + twin|prism (D5/D2)	**Yes**	unflagged (1.0)	0.1340	0.2537	**0.1340**	**0.4000**	**0.2537**	**#1 FDA pre-blot missense_inac** sole 2-generator survivor (twin # prism #8 99.4t aneuploidy, $p=0.0096$) + Bub1b^{H/H} senolytic rescue (PMID:2204831
3	**NIACIN**	it2|it4|it5 (3)	route:R3 + twin|walk (D3)	**Yes**	unflagged (1.0)	0.1920	0.1272	**0.1920**	**0.0480**	**0.1272**	**#1 FDA missense_unst under calibrated** supported across iterations (it2|it4|it5, PMID:24825348
4	**NICOTINAMIDE**	it2|it4|it5 (3)	route:R3 + walk (D3)	**Yes**	unflagged (1.0)	0.1920	0.1272	**0.1920**	**0.0480**	**0.1272**	**Co-#1 FDA missense_unst under calibrated** SIRT2 cofactor precursor (it2|it4|it5, PMID:24825348
5	**GLICLAZIDE**	it5 (1)	route:R2 (D3 Gygi-10k proteome)	**Yes**	unflagged (1.0)	**0.7000**	**0.4638**	**0.1750**	**0.0438**	**0.1159**	**#1 uncalibrated (u=0.7000)**: rais endogenous Bubl $+1.47\;\log_2\te ($2.77\times$, $q=1.1\times 10^ in HCT116 (entr 160)
6	**MIGALASTAT**	it5 (1)	route:R2 (D3 Gygi-10k chaperone)	**Yes**	unflagged (1.0)	**0.7000**	**0.4638**	**0.1750**	**0.0438**	**0.1159**	**Co-#1 uncalibrat (u=0.7000)**: ora pharmacological chaperone raising BubR1 $+1.44\;\log_2\te ($2.72\times$, $q=3.1\times 10^ entry 160)
7	**RAPAMYCIN**	it2|it5 (2)	route:R3 (D5 mTOR / SASP)	**Yes**	unflagged (1.0)	0.1280	0.1280	0.1280	0.1280	0.1280	Dampens mTORC1/SASP i checkpoint-deficie aneuploid models (PMID:2754933 entry 144)
8	**QUERCETIN**	it5 (1)	route:R3 (D5 senolytic pair)	No	unflagged (1.0)	0.1120	0.1120	0.1120	0.1120	0.1120	Paired with dasatinib (D+ senolytic clearanc protocols (entry
9	**METFORMIN**	it2|it5 (2)	route:R3 (D2/D5 AMPK)	**Yes**	unflagged (1.0)	0.0960	0.0960	0.0960	0.0960	0.0960	Iteration 1 & 2 me stress modulator; neutral in PRISM (entry 151)
10	**ATALUREN**	it2|it5 (2)	route:R1 (D1 PTC readthrough)	No	unflagged (1.0)	0.0990	0.0990	0.0990	0.0990	0.0990	Dominated by D3 across all states (R1_score = 0 EVPPI(readthr = 0.0000, entr 159, 162)
D4-1	**SIMVASTATIN**	it2 (1)	**D4 (tumour_context RMS)**	**Yes**	caution_uncalled_pgx (0.7)	*(D4)*	*(D4)*	*(D4)*	*(D4)*	*(D4)*	**Restored Iteratic Rung-5 RMS lea** Phase I paediatric NCT02390843 + human RMS apo via HMGCR (PMID:4124973 gate on SLCO1B1
D4-2	**LOVASTATIN**	it2 (1)	**D4 (tumour_context RMS)**	**Yes**	caution_uncalled_pgx (0.7)	*(D4)*	*(D4)*	*(D4)*	*(D4)*	*(D4)*	**Restored Iteratic Rung-5 RMS lea** validated with

Rank (R5)	Drug	Iterations (count)	Primary route / role	FDA (Phase 4)	Safety status (M_safety)	R4 u_unst	R4 EU	R5 u_unst	R5 u_inact	R5 EU	Evidence summa
											simvastatin ir human RMS apop (PMID:4124973 gate on SLCO1B1
K2-Fail	**VINCRISTINE**	it2|it4|it5 (3)	**Active RMS backbone (K2 fail)**	**Yes**	**both_flagged (0.1)**	0.0000	0.0000	0.0000	0.0000	0.0000	**Sole drug failing safety arms**: CYI *1/*3 + OnSIDE HP:0003202 mu atrophy + 57 paec FAERS reports (entries 152,
K1-Fail	**ALISERTIB / BARASERTIB / BAY-1217389 / CFI-402257 / EMPESERTIB**	it2|it4|it5 (3)	**Do-Not K1 (A1 SAC/Aurora inhibitors)**	No	**avoid (0.0)**	0.0000	0.0000	0.0000	0.0000	0.0000	**Strictly contraindicated**: Mps1/Aurora inhil are *less* effective high-aneuploidy li ($\delta = -0.139, p=0.0096$, entr 151)

(All 52 rows — including the remaining 22 exploratory Route R2 Gygi-10k perturbations, amlexanox, caffeine, fomepizole, cefpirome, fentiazac, paclitaxel, docetaxel, bortezomib, and atorvastatin — are recorded with full provenance in ressources/track2/it5/unified/cross_iteration_candidates.tsv.)